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Algorithms and Applications

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TALKS

Macrina Lobo (Georgia Institute of Technology), Ziqi Zhang (Georgia Institute of Technology) and Xiuwei Zhang (Georgia Institute of Technology). *Spatiotemporal cell type deconvolution leveraging tissue structure.*

Abstract. Spot-based spatial transcriptomics (ST) captures aggregated transcriptomic profiles at spatial locations (spots) in tissues. Deconvolution methods are needed to estimate the proportion of each cell type in every spot and usually leverage a single cell transcriptomic reference (scRNA-seq). Though there are an increasing number of experiments that profile multiple adjacent tissue slices, no deconvolution method leverages 3D tissue structure. Some methods utilize the 2D spatial organization assuming neighboring spots are similar, which is not true in heterogeneous environments. Moreover, most methods aggregate reference scRNA-seq profiles of the same cell type, missing subtle cell state variations.

We present SpaDecoder, a parallelized per-spot deconvolution method for multiple neighboring spatial or temporal ST slices that predicts cell type proportions with a matrix factorization-based objective. SpaDecoder uses slice alignment and intermediate slice imputation, per-spot spatio-transcriptomic neighborhood inference, and 3D Gaussian kernel weights to effectively leverage 3D structure and adapt to homogeneous and heterogeneous tissue environments. We model individual scRNA-seq profiles, instead of cell type aggregated, with a learned scRNA-seq cell to cell type association matrix to capture cell state variability.

The mathematical framework of SpaDecoder supports several downstream analyses. It uncovers key cell type regions and changing composition across slices, identifies colocalized cell types, imputes spatial gene expression, and predicts 3D spatio-temporal scRNA-seq cell locations. SpaDecoder outperforms baseline methods on various metrics, datasets, scenarios, and ablations, and yields interpretable biology, showing it effectively harnesses 3D structure and single cell reference profiles to improve deconvolution. SpaDecoder is available at https://github.com/ZhangLabGT/spadecoder}{https://github.com/ZhangLabGT/spadecoder with the preprint at https://www.biorxiv.org/content/10.64898/2026.02.10.705204v1.

Keywords: deconvolution, spatial transcriptomics, 3D tissue, spatiotemporal

Tianyuan Liu (Institute for Integrative Systems Biology (I2SysBio), Spanish National Research Council (CSIC)) and Ana Conesa (Institute for Integrative Systems Biology (I2SysBio), Spanish National Research Council (CSIC)). *SQANTI-epi: Functional and Quality Annotation of Single-Molecule Epigenomic Data.*

Abstract. A rapidly expanding family of single-molecule chromatin profiling technologies now generates per-read epigenomic signals from long-read sequencing: m6A-based methods (Fiber-seq, SAMOSA, SAM-seq), GpC methylation approaches (nanoNOMe, MeSMLR-seq), multi-mark protocols (SMAC-seq), deaminase-based footprinting (DAF-seq, FOODIE), and protein-targeted strategies (DiMeLo-seq). Each uses distinct enzymatic chemistries, yet no framework exists to assess data quality or compare protocols against common reference epigenomes. Without QC, artifacts propagate silently, and there is no principled way to evaluate whether different technologies yield concordant regulatory element maps.

SQANTI-epi is, to our knowledge, the first QC framework for single-molecule epigenomics. It implements a three-level architecture: library-level metrics (Chromatin Signal-to-Noise from base modification autocorrelation, methylation rate, nucleosome framing) flag improperly treated libraries; region-level 53-element feature vectors — including a 9-bin spatial profile analogous to positional k-mer representations — serve as per-region quality descriptors; and cross-annotation against user-supplied reference tracks (ENCODE cCREs, ATAC-seq, histone marks) measures concordance between single-molecule and population-level data, driving positive-unlabeled learning that scores regulatory elements without negative labels.

The protocol-agnostic design provides a common feature model enabling direct cross-protocol comparison against identical references. Implemented in Rust with a streaming parallel architecture, SQANTI-epi processes a 25-fold coverage GM12878 Fiber-seq dataset (4.7M fibers, 78 Gbp, 368.6M inter-nucleosome regions) in 27 minutes on 64 threads. TSS-proximal regions show 4.3-fold higher concordance with reference peaks than intergenic regions (59.9% vs 14.0%) — a diagnostic ratio whose collapse would reveal over-treatment or systematic false-positive calls.

Keywords: single-molecule epigenomics, quality control, Fiber-seq, long-read sequencing, chromatin accessibility, positive-unlabeled learning

Yanmei Dou (Westlake University). *Unravelling genome-wide mosaic microsatellite mutations at single-cell resolution.*

Abstract. Short tandem repeats (STRs), or microsatellites, are highly mutable genomic elements that modulate gene regulations and are implicated in a range of human diseases. However, detecting mosaic STR mutations at single-cell resolution remains challenging due to both technical and biological complexities. To address this, we developed BayesMonSTR, a robust algorithm that enables accurate detection of mosaic STR mutations. Using this tool in single-cell analysis of human tissues, we reveal an accumulation of longer mosaic STR insertions and deletions (indels) in aging mitotic and post-mitotic cells. Strikingly, prefrontal cortex (PFC) neurons accumulate a higher burden of STR mutations than B cells or lung epithelium, with aged neurons exhibiting a particularly pronounced increase in longer STR deletions. These mutations are enriched at transcription start sites (TSSs) and active enhancers of highly expressed genes. Our work establishes a foundation for genome-wide, hypothesis-free discovery of disease-associated mosaic STR mutations and reveals a previously unexplored landscape of mosaic STR variation in development and aging.

Keywords: microsatellites, short tandem repeats, mosaic mutations, single cell, neurodegeneration

Mao-Jan Lin (Johns Hopkins University), Vikram Shivakumar (Johns Hopkins University) and Ben Langmead (Johns Hopkins University). *ImpuT2T: Pangenome-Based Patching for Human Genome Assemblies.*

Abstract. With improvements in long-read sequencing and assembly methods, many high-quality telomere-to-telomere (T2T) assemblies have become available. These assemblies now serve as important reference collections for sequencing data analyses. However, the long-read sequencing required to generate such a high-quality assembly remains expensive, and is often out of reach for many research groups and applications.

Here, we propose ImpuT2T, a tool to scaffold and patch de novo assemblies with a panel of high-quality assemblies. ImpuT2T takes a fragmented assembly as input and uses the pangenome panel to scaffold and fill inter-contig gaps. Unlike single-reference patching methods, ImpuT2T leverages haplotype diversity in a pangenome panel to infer gap sequences consistent with the underlying haplotype structure. Specifically, ImpuT2T uses flanking region homology to infer the most probable gap sequence between two neighboring contigs, producing more contiguous and user-friendly personalized assemblies at lower cost with minimum error.

Benchmark comparison demonstrates that ImpuT2T is highly effective at patching human HiFi assemblies and outperforms existing gap-filling approaches, highlighting the advantages of pangenome-based patching over single-reference methods.

Keywords: genome assembly, pangenome, patching, scaffolding, pangenomics, reference bias

Dohun Yi (Hanyang University) and Jin-Wu Nam (Hanyang Univ.). *ORION: A High-Performance Barcode and UMI Correction Tool for Oxford Nanopore Single-Cell Sequencing.*

Abstract. Oxford Nanopore long-read single-cell RNA sequencing captures full-length transcripts, enabling isoform-level and variant-level analyses within individual cells. However, Nanopore INDEL errors corrupt cell barcodes and UMIs at multiple stages of preprocessing, and existing tools do not fully account for these stage-specific error structures, often discarding reads that could be rescued with more targeted correction. We present ORION, a barcode and UMI correction tool implemented in C that employs stage-specific error modeling across a five-stage disk-based pipeline, integrating statistical frameshift diagnosis, progressive barcode merging, and transcript-aware UMI deduplication. ORION accepts raw FASTQ and produces corrected, tagged FASTQ compatible with existing workflows, supporting multiple 10x Chromium chemistries. We benchmarked ORION against 7 tools across 3 datasets spanning simulated, CRISPR-edited, and primary cell data on both R9.4.1 and R10.4.1 chemistries. ORION achieved the highest read retention at 91.7% compared to 52-86% for other tools, the highest BC F1 of 0.954 and UMI F1 of 0.625, and the fastest throughput, completing 41 million reads in under 10 minutes with roughly 10 GB memory. Downstream gene expression quantification across 31 pipeline configurations showed that ORION's correction quality translated to expression-level improvements, with ORION-based chains achieving the highest correlations to short-read reference.

Keywords: Oxford Nanopore, Single-cell RNA-seq, Cell barcode correction, UMI deduplication, Long-read sequencing

Anirban Chakraborty (Department of Electrical and Computer Engineering, Michigan State University, East Lansing, MI, USA), Cort Thompson (Department of Biomedical Engineering, Michigan State University, East Lansing, MI, USA), Lauren Wade-Kleyn (Neuroscience Program, Michigan State University, East Lansing, MI, USA), Mark Reimers (Neuroscience Program, Michigan State University, East Lansing, MI, USA) and Erin Purcell (Department of ECE, Department of BME, Michigan State University, East Lansing, MI, USA). *SpatiaXen: An Open-Source Interactive Platform for Cell-Type-Resolved Spatial Transcriptomics Analysis.*

Abstract. Background: In-situ spatial transcriptomics technologies, such as the 10x Genomics Xenium platform, enable single-cell-resolution analysis of gene expression while also preserving the spatial organization of tissue. These platforms can provide new insights into cellular responses to diseases and injuries, as well as therapeutic interventions. They generate tens of millions of spatially resolved transcript coordinates per tissue section. This enables new opportunities to investigate spatially organized biological phenomena, such as neurodegenerative processes and tissue responses around implanted biological devices. However, fully leveraging this potential requires complex computational methods that can handle all the unique features of the in situ spatial transcriptomics data, as they are very large, complex datasets. There are also imaging-related spatial artifacts that need to be addressed before downstream analysis. Also, in most cases, we need to relate gene expression patterns to the locations of known tissue structures or regions rather than only assigning them to individual cells. The current analysis pipelines assign transcripts to individual cells using the cell segmentation technique. But in tissues, such as the brain, where the cellular processes are intertwined and most reads come from the dense neuropil rather than the cell bodies, similar segmentation approaches may introduce error and obscure spatial information.

When studying the brain tissue response to implanted microelectrode arrays, devices that interface with the central nervous system to treat various neurological disorders, these issues can come to the fore. It is important to understand how the individual cell types, including neurons, astrocytes, microglia, etc., respond to the implanted devices at the single-cell resolution to improve the design and biocompatibility of the electrodes. However, the existing computational tools are not well suited to the spatial complexity of brain tissue. To address this challenge, we developed SpatiaXen, an open-source interactive computation platform that provides a complementary analytical framework alongside conventional segmentation-based algorithms. SpatiaXen works with the raw transcript coordinates treated as spatial point data. It generates a spatial map of gene expression using kernel density estimation and classifies cell types using a marker gene-based scoring framework based on the spatial clustering of cell type-specific transcripts around nuclei. This density-based approach captures expression from both the bodies of the cells and the processes as a function of distance from the locations selected by the users. In the future, a cell segmentation pipeline will be incorporated as an option, which will allow the users to try both the density-based and segmentation-based approaches within a single software package. Although we developed and tested this software for brain tissues, this approach can be used for spatial transcriptomics analysis of any Xenium data. We implemented SpatiaXen in Python, and it is structured as an interactive desktop application. SpatiaXen processes standard Xenium output files including parquet or CSV transcript tables, nucleus boundary files and user-defined region coordinates within a modular analysis pipeline. Transcript coordinate data are loaded and can be visualized as a spatial map of downsampled transcripts. Then the users can interactively define the region coordinates, which are used as the region of interests for our analysis. We have also included an option to store the data for reproducibility. This software estimates the gene expression at each spatial location by computing the weighted averages of nearby transcript counts. Total transcript density is estimated on a coarse grid (40 μm spacing, 80 μm bandwidth) to capture spatial variations. Gene specific densities are calculated on a finer grid (5 μm spacing and 7 μm bandwidth). To reduce the spatial artifacts in the data, the gene-specific densities are normalized by the local total transcript density relative to the average total read density. This approach corrects the systematic differences in the read density, including periodic 30–40% variations caused by the field-of-view stitching, without altering the true biological variation. Cell type inference is done using user-defined marker gene sets for each cell type of interest. High-resolution density maps (2.5 μm grid, 3.5 μm bandwidth) were constructed for each marker gene and converted the bounded scores between 0 and 1 using a rank-based exponential scaling because of the right-skewed distribution of the sparse markers. Principal component analysis is applied to per-pixel marker scores, and normalized PC1 loadings of marker genes are used as weights to compute the composite cell type scores. Each nucleus is assigned to the cell type with the highest total score, with confidence defined by the difference between the top scores, allowing ambiguous assignments to be excluded. For studies related to spatial perturbations, including devices, lesions, or tumors, our software measures the distance of each transcript from a user-defined reference surface. The expression values for each cell type are grouped by distance. It enables the analysis of spatial expression patterns and comparison between tissue close to and far from the perturbation. We used ANOVA to evaluate statistical significance with Benjamini-Hochberg false discovery rate corrections at the biological replicate level.

Results: We applied SpatiaXen to Xenium in situ spatial transcriptomics data from rat motor cortex implanted with silicon and polyimide electrodes of 10 and 100 μm cross-sectional sizes at 1 and 6 weeks post-implantation [1]. The software's cell-type inference identified excitatory neurons, interneurons, astrocytes, oligodendrocytes, and microglia. Our analysis indicates that implants are associated with increased inflammatory gene expression in glial cells, accompanied by damage-related reductions in synaptic transmission and activation of protective responses, including remyelination and antioxidant production to preserve the nearby neurons. While our bulk tissue analysis confirmed previously reported glial scar consolidation over time, SpatiaXen's per-cell analysis discovered a progressive increase in inflammatory gene expression (GFAP, Serping1, Gpnmb) within individual reactive astrocytes between 1 and 6 weeks. Using the software's distance analysis, we identified spatial gradients of gene expression extending from the implant location specifically in microglia (C3, Dock8, and Itgam), astrocytes (Gpnmb and Serping1), oligodendrocytes (Hcn2), and neurons (Kif5a). SpatiaXen revealed that smaller implants were associated with fewer reactive cells, rather than decreased per-cell reactivity, an effect revealed by separate tissue-level and per-cell analysis.

Conclusion: While developed for neural device-tissue interfaces, SpatiaXen is applicable to any in situ spatial transcriptomics study involving complex tissues or spatially structured perturbations, including tumor microenvironments, neurodegeneration, TBI, and tissue-biomaterial interfaces. The interactive modular design of the software allows researchers to adapt marker-gene panels and analysis parameters to their tissue type without requiring programming expertise.

[1] A systematic analysis of brain tissue response to microelectrode material and size with single-cell spatial transcriptomics. Cort Thompson, Anirban Chakraborty et. al. doi: https://doi.org/10.64898/2026.03.12.711361

Keywords: Spatial transcriptomics, Single-cell resolution, Kernel density estimation, Xenium, Software

Arseny Shur (Bar-Ilan University), Ido Tziony (Bar-Ilan University) and Yaron Orenstein (Bar-Ilan University). *10-minimizers: a promising class of constant-space minimizers.*

Abstract. Minimizers are sampling schemes which are ubiquitous in almost any high-throughput

sequencing analysis. Assuming a fixed alphabet of size σ, a minimizer is defined by two positive integers k, w and a linear order ρ on k-mers. A sequence is processed by a sliding window algorithm that chooses in each window of length w + k − 1 its minimal k-mer with respect to ρ. A key characteristic of a minimizer is its density, which is the expected frequency of chosen k-mers among all k-mers in a random infinite σ-ary sequence. Minimizers of smaller density are preferred as they produce smaller samples, which lead to reduced runtime and memory usage in downstream applications. Recent studies developed methods to generate minimizers with optimal and near-optimal densities, but they require to explicitly store k-mer ranks in Ω(2^k) space. While constant-space minimizers exist, and some of them are proven to be asymptotically optimal, no constant-space minimizers was proven to guarantee lower density compared to a random minimizer in the non-asymptotic regime, and many minimizer schemes suffer from long k-mer key-retrieval times due to complex computation.

In this paper, we introduce 10-minimizers, which constitute a class of minimizers with promising properties. First, we prove that for every k > 1 and every w ≥ k − 2, a random 10-minimizer has, on expectation, lower density than a random minimizer. This is the first provable guarantee for a class of minimizers in the non-asymptotic regime. Second, we present spacers, which are particular 10-minimizers combining three desirable properties: they are constant-space, low-density, and have small k-mer key-retrieval time. In terms of density, spacers are competitive to the best known constant-space minimizers; in certain (k, w) regimes they achieve the lowest density among all known (not necessarily constant-space) minimizers. Notably, we are the first to benchmark constant-space minimizers in the time spent for k-mer key retrieval, which is the most fundamental operation in many minimizers-based methods. Our empirical results show that spacers can retrieve k-mer keys in competitive time (a few seconds per genome-size sequence, which is less than required by random minimizers), for all practical values of k and w. We expect 10-minimizers to improve minimizers-based methods, especially those using large window sizes. We also propose the k-mer key-retrieval benchmark as a standard objective for any new minimizer scheme.

Keywords: minimizers, universal hitting set, k-mer

Marjan Hosseini (University of Connecticut), Ella Veiner (University Of Connecticut), Thomas Bergendahl (University Of Connecticut), Tala Yasenpoor (University Of Connecticut), Zane Smith (University of Tennessee), Margaret Staton (University of Tennessee) and Derek Aguiar (University of Connecticut, Institute for Systems Genomics). *pHapCompass: Probabilistic Assembly and Uncertainty Quantification of Polyploid Haplotype Phase.*

Abstract. Motivation:

Computing haplotypes from sequencing data, i.e. haplotype assembly, is an important component of molecular and population genetics problems, including interpreting the effects of genetic variation on complex traits and reconstructing genealogical relationships. Assembling the haplotypes of polyploid genomes remains a significant challenge due to the exponential search space of haplotype phasings and read assignment ambiguity; the latter challenge is particularly difficult for haplotype assemblers since the information contained within the observed sequence reads is often insufficient for unambiguous haplotype assignment in polyploid genomes.

Results:

We present pHapCompass, probabilistic haplotype assembly algorithms for diploid and polyploid genomes that explicitly model and propagate read assignment ambiguity to compute a distribution over polyploid haplotype phasings. We develop graph theoretic algorithms to enable statistical inference and uncertainty quantification despite an exponential space of possible phasings. Since prior work evaluates polyploid haplotype assembly on synthetic genomes that do not reflect the realistic genomic complexity of polyploidy organisms, we develop a computational workflow for simulating genomes and DNA-seq for auto- and allopolyploids. Additionally, we generalize the vector error rate and minimum error correction evaluation criteria for partially phased haplotypes. Benchmarking of pHapCompass and several existing polyploid haplotype assemblers shows that pHapCompass yields competitive performance across varying genomic complexities and polyploid structures while retaining an accurate quantification of phase uncertainty.

Availability and implementation:

The source code for pHapCompass, simulation scripts, and datasets are freely available at https://github.com/bayesomicslab/pHapCompass.

Keywords: haplotype assembly, polyploidy, probabilistic graphical models, uncertainty quantification

Michael Goneos (Department of Systems and Computational Biomedicine, Weill Cornell Medicine), Theodore Nelson (Department of Systems and Computational Biomedicine, Weill Cornell Medicine) and Christopher Mason (Department of Systems and Computational Biomedicine, Weill Cornell Medicine).
Pilot discovery of allele-specific modification QTLs from long-read RNA sequencing.

Abstract. RNA modifications are a major regulatory layer of gene expression, but the field still lacks a robust way to ask a mechanistic question: does genetic variation change RNA modification stoichiometry on one allele but not the other? Short read approaches break the link between haplotype, transcript structure, and modification state, and thus are restricted from allele-specific modification detection. Oxford Nanopore direct RNA sequencing changes that by reading all three features on the same molecule. Here, we present PLASMA (Phased Long-Read Analysis of Splicing and Modification with Alleles), a framework for transcript aware, allele specific modification quantification trait locus (modQTL) discovery that integrates transcript structure, per read modification quantification, and within sample haplotype resolved association testing.

Across 18 samples, PLASMA tested 10,696 transcript site comparisons and identified 94 FDR significant modQTLs. This represents a discovery rate of roughly 0.8% of tested comparisons in a modestly sized cohort. m6A sites dominated the discovery set, accounting for 90 of 94 significant events. Critically, some of the strongest effects were linked to variants outside the canonical DRACH motif window rather than to direct motif rewriting. These results suggest that genetic control of RNA modification extends beyond canonical sequence motifs, even in this limited scale dataset.

PLASMA thus provides a proof of concept path toward studying the genetic architecture of the modified transcriptome at the level where regulation occurs, that is, phased, transcript resolved, and molecule specific.

Keywords: Nanopore sequencing, Epitranscriptomics, RNA modification, Allele-specific analysis, Long-read RNA sequencing

Ayse Keskus (NIH), Tanveer Ahmad (NIH/NCI), Isabel Rodriguez (NIH/NCI), Anton Goretsky (NCI/NIH), Ataberk Donmez (NCI/NIH), Sonam Tulsyan (NCI/NIH), Nicholas Syracuse (NCI/NIH), Michael Dean (NIH/NCI) and Mikhail Kolmogorov (NIH). *Minerva: Breakpoint-Guided Allele-Specific Copy Number Inference in Long-Read Cancer Genomes.*

Abstract. Copy-number alterations (CNAs) and structural rearrangements play a key role in cancer evolution, but resolving allele-specific copy-number structure remains difficult in highly rearranged genomes. Existing methods often rely on segmentation and phasing strategies that do not account for structural variant (SV) breakpoints, leading to smoothing across rearrangements and limited resolution, especially for small or focal events. While long-read sequencing provides accurate breakpoint detection and long-range haplotype information, these signals are typically not used together.

We present Minerva, a method for inferring total and allele-specific copy number by directly incorporating SV information. Minerva uses breakpoint-guided segmentation, defining genomic segments based on SV breakpoints so that copy-number states follow the underlying genome structure. Total copy number is estimated using a mixed integer linear program that combines segment-level coverage with constraints from SV junctions, producing consistent integer copy-number states across the genome. For allele-specific inference, Minerva integrates haplotype-tagged read coverage with B-allele frequency (BAF), weighting each based on how informative it is locally, and uses SV junctions to extend haplotype phase across distant regions.

We evaluated Minerva on the CASTLE long-read cancer cell-line panel using a truth set derived from diploid assemblies of matched normal samples. Across nearly all cell lines, Minerva improves chromosome-scale accuracy, yields more stable purity and ploidy estimates, and increases sensitivity and precision in detecting copy-number events. By incorporating SV breakpoints, Minerva resolves segments smaller than 100 bp and avoids smoothing across rearrangements, enabling more accurate reconstruction of complex genomic structures.

Keywords: Copy number variation (CNV), Structural variants (SV), Allele-specific copy number, Complex genome rearrangements, Long-read sequencing, Cancer genomics

Jacob Tye (University of Utah) and Kendell Clement (University of Utah). *MethylSeg: Context-Aware Methylome Segmentation Enables Robust PMD Detection in Array and WGBS Data.*

Abstract. DNA methylation is a repressive epigenetic mark that regulates gene expression. Partially methylated domains (PMDs) are extended regions that capture large-scale methylation dysregulation and are a hallmark of cancers. Many segmentation algorithms have been developed to identify these domains, but they are limited to whole-genome bisulfite sequencing data, excluding more cost-effective and widely available array-based datasets. Furthermore, current methods lack a consistent definition of features that characterize PMDs.

To address these limitations, we present MethylSeg, a context-aware methylome segmentation framework that enables PMD detection across assay types. MethylSeg refines the definition of PMDs by incorporating additional methylation features, including intermediate methylation percentages. It also improves cross-assay compatibility by employing a three-step approach: emission generation, state assignment, and region construction. To generate biologically meaningful emissions, MethylSeg summarizes CpG methylation across genomic windows, allowing each CpG to be interpreted locally and within its broader genomic context. These context-aware emissions are used to assign CpGs to biologically meaningful states using k-means clustering. Finally, contiguous regions are defined by applying a Hidden Markov Model to smooth CpG state transitions and group adjacent same-state CpGs.

To assess the biological relevance of our methodology, we evaluated PMDs identified by MethylSeg and other established methods using chromatin enrichment and functional annotations analyses. MethylSeg-derived PMDs captured expected epigenetic patterns, including depletion of H3K36me2 within PMDs. Overall, MethylSeg provides a flexible and interpretable framework for PMD identification, with improved cross-assay compatibility and improved biological interpretability compared to existing methods.

Keywords: DNA Methylation, Epigenetics, Cancer epigenetics, Partially Methylated Domains (PMDs), Methylome segmentation, DNA methylation arrays, Whole-genome bisulfite sequencing

Weijian Wang (UCLA) and Jingyi Jessica Li (Fred Hutchinson Cancer Center). *scVarSim: Unified Simulation of Genomic Variants and RNA Editing in Single-Cell RNA-seq Data.*

Abstract. Single-cell RNA sequencing captures genetic variants and post-transcriptional RNA editing events alongside gene expression, enabling cell-type-specific sequence-level analyses. However, accurate detection of single-nucleotide polymorphisms (SNPs), indels, and RNA editing events remains challenging due to sparse coverage and strong confounding between genomic polymorphisms and RNA editing. Existing benchmarking of variant calling and RNA editing detection lacks realistic simulations with explicit single-cell ground truth. To fill this gap, we present scVarSim, a simulation framework trained on real scRNA-seq data that generates realistic synthetic reads with ground truth for SNPs, indels, and RNA editing. scVarSim preserves key empirical properties of single-cell data, including gene expression distributions, genome-wide coverage profiles, site-specific allelic ratios, and RNA editing levels. Using scVarSim, we systematically benchmarked four RNA editing callers and four variant callers across three aligners, identifying JACUSA2 and DeepVariant as the most robust methods, respectively, and showing that SNPs and RNA editing act as strong mutual confounders. Across all methods, inference at single-cell resolution is sensitivity-limited, with quantitative estimates becoming accurate only conditional on successful detection. Together, these results establish practical guidelines for sequence-level analysis of scRNA-seq data and provide a testbed for method development.

Keywords: scRNA-Seq, Simulation Framework, Variant Calling, RNA Editing, Benchmark

POSTERS

Rohit Khurana (Perelman School of Medicine, University of Pennsylvania), Yichen Henry Liu (Department of Computer Science, Vanderbilt University) and Xin Maizie Zhou (Vanderbilt University). *DipSVFilter: filtering false-positive structural variants through cluster-aware diploid hypothesis modeling.*

Abstract. False positives remain a major limitation of long-read structural variant (SV) callsets, especially in complex loci containing clustered, overlapping, or interacting SV candidates. Such errors reduce the reliability of downstream analyses and make post-calling refinement an important bottleneck for accurate SV discovery. Existing SV filters typically evaluate candidate SVs independently using local alignment-derived features, without explicitly modeling alternative diploid sequence explanations for clustered SV signals.

We present DipSVFilter, a cluster-aware post-calling SV filtering framework for long-read data. DipSVFilter first groups nearby candidate SVs into local clusters. Within each cluster, it constructs alternative pseudo-contigs by inserting candidate SVs into the reference sequence and enumerates possible diploid pseudo-contig combinations. It then prunes implausible configurations by comparing candidate-derived k-mers with k-mers observed in reads aligned to the local genomic region, excluding hypotheses that imply unsupported sequence content. For the remaining hypotheses, DipSVFilter realigns local reads to candidate diploid sequences and applies a time-distributed CNN-Transformer classifier to quantify residual SV signal after realignment. The best-supported diploid configuration is selected by minimizing residual SV signal, and the implied SVs are retained as the filtered SV set.

Initial experiments suggest that DipSVFilter improves F1 on complex SV clusters relative to existing SV filtering methods, including CSV-Filter and Samplot-ML, while reducing false positives in clustered loci. These results highlight the value of jointly modeling clustered SV candidates as alternative diploid sequence hypotheses and suggest a practical strategy for improving the reliability of long-read SV callsets in structurally complex regions.

Keywords: Structural variants, Long-read sequencing, SV filtering, Complex genomic loci

Xiaoyu Zhao (University of California San Diego), Isabella Panagiotou (University of California San Diego), Francisco Sánchez-Rivera (Massachusetts Institute of Technology), Marcus Kelly (University of California San Diego) and Trey Ideker (University of California San Diego). *Massively parallel analysis of coding mutations with CodonPrime.*

Abstract. Most coding mutations are of unknown function, limiting their biological and therapeutic interpretation. While prime editing enables precise genomic alterations, mutations often reside in sequence contexts unfavorable to current prime-editing guide RNA (pegRNA) designs. Here we develop CodonPrime, a prime-editing framework that exploits codon degeneracy surrounding a mutation site to substantially expand its targeting set of pegRNAs. We use CodonPrime to screen ~2500 amino acid mutations spanning 298 cancer-associated genes, yielding an 11.4-fold increase in per-mutation editing efficiency relative to conventional pegRNA designs. Approximately 10% of these mutations enhance cell growth, validating known oncogenic hotspots and uncovering 30 previously uncharacterized mutations with oncogenic potential (e.g. HNF1AR272C). Our analysis uncovers dominant effects in paralogous genes (e.g. RHOBTB1/2) and organization of mutant phenotypes in functional pathways (e.g. PI3K-MTOR signaling). We formulate a general-purpose transformer for design of codon-optimized pegRNAs, enabling scalable interrogation of human coding variants.

Keywords: Prime editing, Codon optimization, pegRNA design, Functional variant screening, Pathway-level phenotypes

Betselot Zerihun (Ethiopian Public Health Institute), Chala Bashea (Ethiopian Public Health Institute), Ayinalem Alemu (Ethiopian Public Health Institute), Gemechu Tadesse (Ethiopian Public Health Institute), Feyisa Chala (Ethiopian Public Health Institute), Alemayehu Godana (Addis Ababa University) and Helen Nigussie (Addis Ababa University). *Nationwide Computational Genomic Characterization of Drug Resistance and Lineage Structure in Ethiopian Mycobacterium tuberculosis.*

Abstract. Tuberculosis (TB) remains a global health challenge; we applied bioinformatics to study lineage diversity and drug resistance in Ethiopian Mycobacterium tuberculosis. A bioinformatics pipeline was applied to 1,888 Ethiopian MTB whole-genome datasets (2008-2026) from NCBI, including quality control with Fastp, mapping to the H37Rv reference, variant calling with Snippy, lineage and resistance profiling with TB-Profiler, phylogenetic reconstruction using IQ-TREE, and statistical analyses in Python.

Among the datasets, 66.3% were drug-sensitive (DS-TB), 17.3% multidrug-resistant (MDR-TB), 2.7% pre-extensively drug-resistant (pre-XDR-TB), and 0.5% extensively drug-resistant (XDR-TB). Resistance was highest for isoniazid (26.0%) and streptomycin (23.9%). Recent increases in bedaquiline and clofazimine resistance were observed (1.2%), with combined delamanid/pretomanid resistance in 0.9% of cases; no resistance to linezolid or cycloserine was detected. Predominant mutations included rpoB p.Ser450Leu (72.7%) and katG p.Ser315Thr (89.2%), while fluoroquinolone resistance was driven by gyrA p.Ala90Val (~42%) and delamanid/pretomanid resistance by ddn c.2547_2565dup (82.4%). rv0678 c.139dupG mutations linked to bedaquiline/clofazimine were detected before 2018, with rising prevalence during the pre-implementation period.

Lineage 4, particularly sub-lineage 4.2.2.2 (28.7%), predominated, whereas Lineage 3 (3.1.1) showed a higher MDR-TB burden. Lineages 7 and 1 were largely drug-sensitive. MTB lineages were significantly associated with resistance profiles ($p < 0.001$; Cramer's $V = 0.26$), with CAS1-Delhi and Lineage 3 showing lower odds of resistance (OR = 0.61 and 0.83). Migration-related patterns included the first detection of Lineage 9 in a refugee.

This study highlights a complex, lineage-specific drug-resistance landscape in Ethiopia, emphasizing the need for strengthened genomic surveillance and targeted TB control strategies.

Keywords: Drug-resistance, Ethiopia, Lineages, MDR-TB, Mycobacterium tuberculosis, phylogenomics, pre-XDR-TB

Euijeong Sung (Department of Biotechnology, College of Life Science and Biotechnology, Yonsei University), Ilseok Choi (Department of Biotechnology, College of Life Science and Biotechnology, Yonsei University), Insuk Lee (Department of Biotechnology, College of Life Science and Biotechnology, Yonsei University) and Sungsoo Ahn (Division of Rheumatology, Yonsei University College of Medicine, Yongin, South Korea). *Single-Cell RNA Sequencing Uncovers Divergent Interferon Programs between Anti-TNFα Responders and Non-Responders.*

Abstract. Many patients with rheumatoid arthritis (RA) do not benefit from tumor necrosis factor-α (TNF-α) inhibitors, yet the molecular basis of this variability remains unclear. To explore cell-type-specific signals associated with treatment response, we profiled peripheral blood mononuclear cells from RA patients before therapy using single-cell RNA sequencing. After applying standard preprocessing steps, including dropout correction and aggregation strategies to improve data robustness, we obtained expression profiles suitable for downstream analysis. We then constructed cell-type-resolved co-expression maps and compared transcriptional programs between responders and non-responders.

This comparison revealed a clear separation along the interferon axis. Responders showed broad activation of interferon-α–stimulated genes across multiple immune cell types, with particularly strong signals in classical monocytes. In contrast, non-responders exhibited interferon-γ–driven signatures, mainly in NK and T cell populations. Perturbation-based analysis further supported the association between interferon-α activity and favorable treatment outcomes, whereas interferon-γ signaling was linked to poor response. These patterns were reproducible across preprocessing strategies and remained consistent after accounting for disease activity.

Taken together, our results indicate that differential engagement of type I and type II interferon pathways contributes to variability in response to TNF-α blockade in RA. Interferon-related expression signatures may therefore provide a useful basis for patient stratification and offer insight into potential therapeutic strategies.

Keywords: single-cell RNA sequencing, rheumatoid arthritis, treatment response, immune cell heterogeneity

Sehwa Hong (Research Institute, National Cancer Center; Interdisciplinary Program for Bioinformatics, Seoul National University), Daejin Hyung (Research Institute, National Cancer Center), Hyeyeong Hwang (Research Institute, National Cancer Center), Yeonhui Lee (Research Institute, National Cancer Center), Soo Young Cho (Department of Molecular and Life Science, Hanyang University), Je Kyung Seong (Korea Model Animal Priority Center, Research Institute for Veterinary Science, Seoul National University) and Charny Park (Research Institute, National Cancer Center). *Long-Read Single-Cell RNA Sequencing Reveals Isoform-Level Transcriptional and Immune Repertoire Diversity in Influenza Infection.*

Abstract. Single-cell RNA sequencing (scRNAseq) has advanced our understanding of molecular mechanisms underlying immune responses to viral infection; however, its reliance on short-read sequencing limits the characterization of isoform-level regulation and accurate reconstruction of immune receptor diversity. In this study, we performed PacBio long-read scRNAseq on lung tissues at multiple time points (days 0, 8, and 28 post-infection), enabling full-length transcriptome analysis and immune repertoire reconstruction simultaneously at single-cell resolution.

Our analysis reveals that influenza infection triggers a widespread, cell-type-specific splicing program that is often invisible at the gene level. We identified splicing changes in genes involved in innate immune function and host-pathogen interactions, including a retained intron in Cd14 and a domain truncation in Dnm2, which may modulate macrophage signaling and viral entry efficiency during infection. Long-read sequencing also enabled the reconstruction of a paired, fully covered immune receptor repertoire, uncovering a distinct class-switch recombination toward IgG2a and IgG3 isotypes following infection, along with clonal expansion of specific VH subgroups such as IGHV1-14. Somatic hypermutation (SHM) patterns further confirmed plasma cell maturation during the post-infection phase.

Together, our study demonstrates that long-read scRNAseq overcomes key limitations of conventional approaches by enabling simultaneous resolution of isoform diversity and immune receptor architecture, providing new insights into the molecular and clonal mechanisms underlying host antiviral immune responses. These findings also highlight the potential of this approach for identifying novel diagnostic biomarkers and therapeutic targets by linking transcriptomic structural diversity with adaptive immune responses.

Keywords: Influenza, Long-read single-cell RNAseq, Alternative isoforms, Immune Repertoire

Jonathan Mandl (Bar-Ilan University) and Yaron Orenstein (Bar-Ilan University). *PrimerDesigner: Designing efficient primers for synthesizing large protein libraries without cross-hybridization.*

Abstract. Efficient synthesis of large protein libraries is key to making high-throughput biochemistry scalable. Advances in DNA microarray technology now allow more than 170,000 designed oligonucleotides to be synthesized in a single array, dramatically reducing cost and enabling pooled amplification for downstream mutagenesis reactions. Effective amplification of short oligonucleotides into long coding sequences requires efficient primer design without cross-hybridization. However, no primer-design method currently guarantees complete coverage of the protein-coding sequences with maximum efficiency and no cross-hybridization over multiple proteins or multiple variants of the same protein. Here, we present PrimerDesigner, a suite of methods to design the most efficient primers for synthesizing large protein libraries with complete protein-coding sequence coverage and without cross-hybridization. We first prove that primer design with complete coverage and without cross-hybridization is NP-hard, even for a single protein. As a solution, we formulate the primer-design problem as an integer linear program (ILP). When no cross-hybridization is possible, we solve the primer-design problem for a single protein in linear time. Moreover, we extend our ILP formulation to handle multiple variants of the same protein. We demonstrate that the methods in PrimerDesigner produce an optimal solution in only a few minutes for a single protein and within a couple of hours for multiple proteins while achieving higher efficiency than a greedy baseline approach. We expect PrimerDesigner to enable the synthesis of large protein libraries at an unprecedented scale and efficiency.

Keywords: primer design, integer linear programming, NP-hard

Manfei Bella Xie (Department of Biomedical Engineering, Vanderbilt University), Yuling Zhu (Department of Biomedical Engineering, Vanderbilt University), Yunfei Hu (Department of Computer Science, Vanderbilt University), Yiru Chen (Systems and Informatics of Zhejiang University-University of Edinburgh Institute), Bingshan Li (Department of Molecular Physiology Biophysics, Vanderbilt University), Shan Meltzer (Department of Pharmacology, Vanderbilt University), Zixuan Cang (Department of Mathematics and Center for Research in Scientific Computation, North Carolina State University), Yanxiang Zhao (Department of Mathematics, The George Washington University), Lu Zhang (Department of Computer Science, Hong Kong Baptist University) and Xin Maizie Zhou (Department of Biomedical Engineering, Computer Science, and Data Science Institute, Vanderbilt University). *3DComFlow: An Optimal Transport Framework that Deciphers Intra- and Intercellular Signaling Landscapes in 3D Tissue Systems.*

Abstract. Cell–cell communication (CCC) is fundamental to multicellular life, governing development, tissue homeostasis, and immune responses. Disease often arises when cells fail to communicate properly or misinterpret molecular signals. Understanding how CCC pathways are disrupted can therefore reveal new therapeutic opportunities. Recent advances in single-cell RNA sequencing (scRNA-seq) have transformed CCC studies from labor-intensive, low-throughput experimental approaches into scalable, data-driven analyses capable of probing diverse cell types and signaling interactions. However, scRNA-seq inherently lacks spatial context, despite the fact that most CCC events occur over short distances. Spatial transcriptomics (ST) offers a promising alternative, yet CCC inference from ST data remains underexplored—particularly in jointly modeling inter- and intracellular signaling and accounting for ligand–receptor competition. To address this gap, we propose a novel optimal transport (OT)-based framework, ComFlow, that captures complex CCC dynamics, including global competition and cooperation among signaling molecules within and between cells or spatial spots. Furthermore, CCC in three-dimensional (3D) tissue structures is largely unexplored. We develop a robust workflow, 3DComFlow, that integrates state-of-the-art alignment methods to map multiple tissue slices from different coordinate systems into a unified 3D space, enabling CCC inference beyond traditional 2D analyses. This approach reveals spatial communication patterns that are otherwise undetectable. We apply our framework across diverse datasets spanning species (human and mouse), varying cell numbers, gene panels, and biological conditions. Comparative analyses (e.g., disease vs. control) uncover both conserved and context-specific signaling networks, demonstrating the broad utility of our approach for advancing spatial systems biology and precision medicine.

Keywords: Spatial Transcriptomics, Integration, Alignment, 3D Reconstruction, Cell-cell Communication, Intercellular Signal, Intracellular Signal, Optimal Transport

Pan Liu (Fred Hutchinson Cancer Center) and Jingyi Jessica Li (Fred Hutchinson Cancer Center / University of Washington). *mcRigor: a statistical method to enhance the rigor of metacell partitioning in single-cell data analysis.*

Abstract. In single-cell data analysis, addressing sparsity often involves aggregating the profiles of homogeneous single cells into metacells. However, existing metacell partitioning methods lack checks on the homogeneity assumption and may aggregate heterogeneous single cells, potentially biasing downstream analysis and leading to spurious discoveries. To fill this gap, we introduce mcRigor, a statistical method to detect dubious metacells, which are composed of heterogeneous single cells, and optimize the hyperparameter(s) of a metacell partitioning method. The core of mcRigor is a feature-correlation-based statistic that measures the heterogeneity of a metacell, with its null distribution derived from a double permutation scheme. As an optimizer for existing metacell partitioning methods, mcRigor has been shown to improve the reliability of discoveries in single-cell RNA-seq and multiome (RNA + ATAC) data analyses, such as uncovering differential gene co-expression modules, enhancer-gene associations, and gene temporal expression. Moreover, mcRigor enables benchmarking and selection of the most suitable metacell partitioning method with optimized hyperparameter(s) tailored to a specific dataset, ensuring reliable downstream analysis. Our results indicate that among existing metacell partitioning methods, MetaCell and SEACells consistently outperform MetaCell2 and SuperCell, albeit with the trade-off of longer runtimes.

Keywords: single-cell data, metacell, homogeneity, rigor, optimization

Jia-Jun Liu (University of Pittsburgh), Qingqi Lin (University of Pittsburgh), Wenjia Wang (University of Pittsburgh), Mengying Xia (University of Pittsburgh), Jian-Hua Luo (University of Pittsburgh), Sungjin Ko (University of Pittsburgh) and Silvia Liu (University of Pittsburgh). *Characterizing isoform expression and mutation using bulk and single-cell long-read RNA sequencing.*

Abstract. Long-read RNA-seq has emerged as a powerful technology for characterizing transcriptome complexity through the capture of full-length transcripts. Compared with short-read sequencing, long-read platforms provide the opportunity to quantify isoform-specific expression and to study transcript-level variations that are critical for understanding gene regulation and disease mechanisms. However, existing computational tools typically do not emphasize the annotation of individual long reads, and truncated long reads with multiple possible alignments are frequently discarded or inaccurately quantified, leading to potential bias in downstream analyses. Moreover, only a limited number of computational approaches have been developed to quantify RNA mutation and editing events at isoform resolution using long-read data.

To address these challenges, we developed a novel computational approach for the comprehensive analysis of long-read RNA-seq data. Our algorithm first annotates individual long reads with all the possible isoform assignments. Next, it estimates the expression level of known isoform, alternative splicing, and fusion transcripts by explicitly modeling ambiguous read assignments, allowing truncated or multi-mapped reads to contribute probabilistically to transcript quantification. Building upon this framework, the second stage of the method estimates RNA mutation and editing expression at the isoform level, enabling integrated characterization of transcript structure and sequence variation. The proposed algorithm was systematically evaluated using multiple simulated and real long-read RNA-seq datasets. The results demonstrate improved accuracy in transcript quantification and robust detection of RNA mutation events. Importantly, the method is compatible with both bulk and single-cell long-read RNA-seq data. Overall, this work provides a generalizable computational tool that facilitates comprehensive isoform-resolved transcriptome analysis for a wide range of biomedical applications.

Keywords: long-read RNA-seq, isoform expression, isoform mutation, RNA editing

Chenxin Flora Jiang (University of California, Los Angeles), Changhu Wang (Fred Hutchinson Cancer Center) and Jingyi Jessica Li (Fred Hutchinson Cancer Center / University of Washington). *Nullstrap-DE: A General Framework for Calibrating FDR and Preserving Power in DE Methods, with Applications to DESeq2 and edgeR.*

Abstract. Differential expression (DE) analysis is a key task in RNA-seq studies, aiming to identify genes with expression differences across conditions. A central challenge is balancing false discovery rate (FDR) control with statistical power. Parametric methods such as DESeq2 and edgeR achieve high power by modeling gene-level counts using negative binomial distributions and applying empirical Bayes shrinkage. However, these methods may suffer from FDR inflation when model assumptions are mildly violated, especially in large-sample settings. In contrast, non-parametric tests like Wilcoxon offer more robust FDR control but often lack power and do not support covariate adjustment. We propose Nullstrap-DE, a general add-on framework that combines the strengths of both approaches. Designed to augment tools like DESeq2 and edgeR, Nullstrap-DE calibrates FDR while preserving power, without modifying the original method's implementation. It generates synthetic null data from a model fitted under the gene-specific null (no DE), applies the same test statistic to both observed and synthetic data, and derives a threshold that satisfies the target FDR level. We show theoretically that Nullstrap-DE asymptotically controls FDR while maintaining power consistency. Simulations confirm that it achieves reliable FDR control and high power across diverse settings, where DESeq2, edgeR, or Wilcoxon often show inflated FDR or low power. Applications to real datasets show that Nullstrap-DE enhances statistical rigor and identifies biologically meaningful genes.

Keywords: differential expression, false discovery rate, calibration

Suhani Balachandran (NIH / NHGRI), Steven Solar (NIH / NHGRI) and Adam Phillippy (NIH / NHGRI).
Measuring the copy number and active fraction of 5S and 45S rDNA units using long-read nanopore sequencing.

Abstract. Human ribosomal RNA genes are organized as tandemly repeating arrays of a composite repeat structure known as ribosomal DNA (rDNA), with the 5S rDNA array located on Chromosome 1 and multiple 45S rDNA arrays located on the short arms of acrocentric chromosomes. Each genome contains hundreds of rDNA copies, and their copy number and transcriptional activity can vary widely between individuals, making them difficult to analyze from sequencing data alone. To enable the analysis of these important genes and uncover any associations with disease, we developed a method that can estimate rDNA copy number and activity using long-read nanopore sequencing. Rather than mapping reads to a reference genome, we instead identify individual rDNAs directly on the single-molecule sequencing reads and infer their transcriptional activity based on the observed methylation patterns. Application of our pipeline to thousands of brain- and blood-derived nanopore samples from the NIH Center for Alzheimer's and Related Dementias showed a weak positive correlation between the number of active 5S and 45S copies, no correlation between the total number of 5S and 45S copies, and a relative consistent number of active rDNA units regardless of sample age or total rDNA copy number. We also detected lower than expected sequencing coverage across the rDNAs, with the degree of bias, as well as the estimated array size itself, varying by sequencing platform. Our work provides a new sequencing-based method for measuring rDNA copy number and transcriptional activity, helping to understand these less well characterized yet important regions of the genome.

Keywords: rDNA, copy number, methylation, long-read nanopore sequencing, sequencing bias

Gabriel Silva (Auburn University College of Forestry Wildlife and Environment), Temitope Folorunso (Auburn University College of Forestry Wildlife and Environment), Lori Eckhardt (Auburn University College of Forestry Wildlife and Environment) and Janna Willoughby (Auburn University College of Forestry Wildlife and Environment). *Benchmarking the Interaction Between Sequencing Depth, Algorithm Choice, and Genome Architecture in Fungal Genome Assembly.*

Abstract. High-quality genomes are essential for comparative, functional, and evolutionary studies, yet although fungal genomes vary widely in size, structural rearrangements, and gene architecture, the effects of assembly algorithms and sequencing strategies on fungal genome reconstruction remain little explored. Most assembly benchmarks have focused on bacterial genomes, leaving uncertainty about how sequencing depth, assembly algorithm, and genome structure interact in fungi. Here, we benchmarked fungal genome assembly performance using both simulated and empirical short- and long-read sequencing datasets. Simulated reads were generated from 66 complete fungal genomes spanning broad ranges of genome size, GC content, and gene density. We evaluated short-read, long-read, hybrid, and long-read-plus-polishing workflows across sequencing depths from 10x to 100x and validated key trends using sequencing data from 10 field-collected fungal isolates. Assembly performance depended primarily on interactions between sequencing depth, assembler choice, and genome architecture. Long-read assemblies showed the largest improvements in contiguity and completeness, with most gains achieved by ~20-40x and limited improvement beyond this threshold. Short-read polishing substantially reduced base-level errors at low additional cost, with ~10-20x short-read coverage generally sufficient to approach maximal error reduction. Hybrid assemblers exhibited strong trade-offs between scaffold continuity, nucleotide accuracy, and computational demand. Furthermore, larger and more feature-rich genomes required greater sequencing depth and computational resources, whereas GC content had minor effects. Across both simulated and empirical datasets, Flye followed by Polypolish consistently provided one of the best balances among contiguity, accuracy, completeness, and computational efficiency.

Keywords: Fungal genome assembly, Genome assembly benchmarking, Long-read sequencing, Short-read sequencing, Hybrid assembly, Comparative genomics, Computational genomics

Jacqueline Durney (University of Utah). *Loss of local methylation coordination underlies structured intermediate methylation states in colorectal cancer.*

Abstract. DNA methylation dysregulation is a hallmark of colorectal cancer (CRC), but how epigenetic instability is structured across individual cells and DNA molecules remains poorly understood. Here, we use single-cell bisulfite sequencing to characterize DNA methylation at both cellular and molecular resolution in primary CRC and matched normal colon tissue. We find that cancer-associated methylation changes extend beyond shifts in average methylation levels to include a pervasive loss of local coordination between neighboring CpGs. Across CpG islands, gene promoters, and large genomic domains, tumor cells exhibit increased methylation discordance alongside reduced methylation haplotype load, indicating disruption of coordinated methylation maintenance. By leveraging read-level metrics of methylation heterogeneity, we distinguish between intercellular variability and intramolecular discordance and show that intermediate methylation states in CRC primarily arise from mixtures of fully methylated and unmethylated cells rather than stable partial methylation within individual cells. Notably, observed methylation variability is significantly lower than expected under stochastic models, supporting a coordinated, non-random basis for epigenetic dysregulation. We further demonstrate that patient-specific methylation programs account for the majority of epigenetic variance and are conserved across tumor sites, while genomic regions that transition to intermediate methylation states exhibit elevated baseline variability in normal tissue, suggesting intrinsic susceptibility to dysregulation. Together, these results reveal that DNA methylation changes in colorectal cancer reflect a structured breakdown of coordinated epigenetic maintenance shaped by both patient-specific programs and region-specific properties, rather than purely stochastic processes.

Keywords: single-cell, methylation, colon cancer

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Abstract. Background

High-grade serous ovarian carcinoma (HGSC) is an aggressive malignancy with high recurrence rates and lim-ited targeted treatment options. Bulk transcriptomic subtypes are being investigated to stratify HGSC tumors but are interpreted as reflecting malignant-intrinsic transcriptional states. Because bulk tumors comprise mix-tures of malignant, stromal, and immune cells, it remains unclear whether subtype structure reflects tumor cell state or cellular composition, or both.

Approach. We quantified contributions of cellular composition and tumor-intrinsic expression to HGSC subtype identity. We combined 1,900 single-cell-informed pseudobulk simulations, varying epithelial, stromal, and immune fractions, with deconvolution-based modeling of 1,834 primary HGSC tumors across eight cohorts.

Results. In simulated tumors, subtype probabilities were determined by cellular composition alone, with MES increasing with fibroblast abundance and IMR/PRO varying along an opposing immune axis. These dependencies were reproduced in primary tumors, where inferred cell-type proportions associated with subtype probabilities and aligned with the principal axis separating subtypes. Subtype separation aligned with PC2 (η^2=0.463; ~46% vari-ance) but not PC1 (η^2=0.005; <0.1% variance) and was reduced after adjusting for composition. Regression models confirmed that composition explained the majority of subtype variance, with the strongest effect for MES. Composition-based models achieved AUCs of 0.81-0.95, while PC-based expression features improved prediction (ΔAUC 0.03-0.14), but did not redefine the primary subtype axis.

Conclusions

HGSC transcriptomic subtypes are largely explained by tumor ecosystem composition rather than stable tumor-intrinsic states. This framework enables disentangling compositional and intrinsic signals in bulk tumor transcriptomics.

Keywords: Bulk RNA-seq deconvolution, Cellular composition modeling, Single-cell-informed simulation, Cancer transcriptomics

Aley Abdel-Ghaffar (Brown University). *Understanding Clinical and Genetic Features Associated with COVID-19 Disease Severity in Rhode Island.*

Abstract. The rapid emergence of SARS-CoV-2 variants has highlighted the need to understand how clinical and viral genetic diversity jointly influence disease severity. While host factors such as age, sex, and comorbidities are well-established determinants of COVID-19 outcomes, the contribution of viral genetic variation remains less characterized, partly due to the methodological challenges of population structure in viral genomics.

Using a dataset of more than 11,000 SARS-CoV-2 sequences with linked clinical metadata provided by the Rhode Island Department of Health, we performed an integrated analysis of clinical and viral genetic determinants of hospitalization risk spanning March 2020 to August 2024. We first applied recursive feature elimination with cross-validation and permutation-based feature importance to identify key clinical predictors, finding that age, sex, vaccination status, and specific comorbidities, particularly cancer, diabetes, obesity, and lung disease, were the strongest predictors of severe disease.

To identify viral genetic associations, we conducted a genome-wide association study correcting for clinical covariates and kinship using linear mixed models. To address viral population structure, samples were stratified into six genetic clusters via principal component analysis. An overall GWAS identified seven genome-wide significant variants; however, all were confirmed as Delta lineage-tagging artifacts concentrated in a single cluster. Cluster-stratified analyses revealed four genuine associations in three lineage groups, including missense variants in the nucleocapsid and ORF6 genes and synonymous variants in ORF1ab. Burden analyses further identified six genomic regions with cumulative rare variant signals, four mapping to ORF1ab.

These results demonstrate that controlling for both clinical covariates and viral population structure is essential for distinguishing true genotype-phenotype associations from lineage confounding in viral GWAS.

Keywords: SARS-CoV-2, COVID-19, genome-wide association study, viral genomics, population structure, disease severity, hospitalization, clinical feature selection, viral genetic variation, lineage stratification, linear mixed models, burden analysis, ORF1ab, nucleocapsid, Rhode Island

Qingqi Lin (University of Pittsburgh), Mengying Xia (University of Pittsburgh), Jia-Jun Liu (University of Pittsburgh), Minwook Kim (University of Pittsburgh), Joseph Kwon (University of Pittsburgh), Jian-Hua Luo (University of Pittsburgh), Sungjin Ko (University of Pittsburgh) and Silvia Liu (University of Pittsburgh). *Spatial and Long-Read Transcriptomics Reveal the Microenvironment in Mixed Liver Tumors.*

Abstract. Mixed liver tumors, particularly combined hepatocellular–cholangiocarcinoma (cHCC-CCA), are highly aggressive malignancies characterized by pronounced cellular heterogeneity and complex tumor microenvironments (TME). A defining feature of these tumors is the coexistence of hepatocytic and biliary cell populations within the same lesion, suggesting intricate cellular interactions and lineage plasticity. However, current transcriptomic approaches are largely limited to gene-level measurements and fail to capture alternative splicing, an important regulatory layer in cancer biology. While spatial transcriptomics preserves tissue architecture, it lacks isoform-level resolution, whereas long-read RNA sequencing resolves full-length transcript isoforms but loses spatial context.

To address this limitation, we analyzed clinical specimens from patients with mixed liver tumors using a dual-technology framework combining imaging-based spatial multi-omics (CosMx) and long-read RNA sequencing (PacBio). We developed a hierarchical Bayesian meta-integration framework to link spatial gene expression with isoform-level transcript information. In this model, latent isoform abundances are inferred from spatial gene expression through a probabilistic gene–isoform mapping constrained by long-read–derived references.

This framework enables the inference of spatially structured isoform usage and cell-type composition across the TME. By integrating transcriptomic, proteomic, and spatial information, we identify spatially variable isoform patterns and their associations with distinct cellular states and microenvironmental contexts. Furthermore, leveraging long-read sequencing, we characterize spatial patterns of mutation and splicing variation across tumor subpopulations.

Together, this study provides a framework for integrating spatial multi-omics with long-read transcriptomics, offering new insights into tumor organization and heterogeneity at the isoform level.

Keywords: Spatial transcriptomics, Long-read RNA sequencing, Isoform analysis, Multi-omics integration

Hao Yuan (Michigan State University), Aishwarya Mandava (University of Colorado Anschutz Medical Campus), Kewalin Samart (University of Colorado Anschutz Medical Campus), Julia Ganz (Michigan State University) and Arjun Krishnan (University of Colorado Anschutz Medical Campus). *Linking Genetic Risk to Disease-Relevant Cellular States via Metacell-Informed Modeling with ICePop.*

Abstract. Genome-wide association studies (GWAS) have identified thousands of loci associated with complex diseases, yet translating these population-level signals into specific cellular contexts remains a major challenge. Integrating GWAS with single-cell transcriptomic data enables systematic identification of disease-relevant cell types, but existing approaches face a fundamental tradeoff. Methods such as seismic achieve high statistical power by operating at the annotated cell-type level, but miss heterogeneous signals confined to specific cellular states. In contrast, single-cell-resolution methods like scDRS capture heterogeneity but often lack power to detect subtle associations. Here, we present ICePop (Informative Cell Populations), a framework that resolves this tradeoff by performing disease-cell association analysis at metacell resolution. By aggregating transcriptionally similar cells, ICePop achieves statistical power comparable to cell-type-level methods while retaining sensitivity to heterogeneous disease signals within cell types. In simulations, ICePop maintains appropriate false positive rates and demonstrates superior power when disease effects are concentrated in cellular subpopulations. Applying ICePop to 81 traits across cell types in Tabula Muris, we identify 2,178 disease-cell type associations, including preferential vulnerability of differentiated gut epithelial cells in ulcerative colitis and immune-stressed lung capillary endothelial states associated with lung capacities. Clustering diseases based on ICePop's metacell association profiles reveals disease groupings that diverge from genetic risk-based clustering, highlighting differences in cellular rather than genetic etiology. In autism spectrum disorder, ICePop implicates specific enteric neuron subtypes, suggesting a role for the enteric nervous system in gastrointestinal comorbidities. Together, ICePop enables resolution of disease-relevant cellular states and supports generation of testable, cell-state-specific hypotheses.

Keywords: Single-cell RNA-seq, Metacell analysis, Disease-cell type association, GWAS integration

Omar Ahmed (Johns Hopkins University), Haowen Zhang (Georgia Institute of Technology), Benjamin Langmead (Johns Hopkins University) and Li Song (Dartmouth College). *Quality Control of Single-Cell ATAC-seq data without peak calling using Chromap.*

Abstract. Single-cell ATAC-seq (scATAC-seq) profiles the heterogeneity of chromatin accessibility across thousands of cells rather than only getting a single, bulk view. The common practice of ATAC-seq data analysis based on features defined by bulk peaks can miss rare cell types in the single-cell setting. This inspires the development of highly efficient methods, such as SnapATAC and ArchR, that use the features defined by the fixed-size genome intervals. However, the skipping of peak calling disenables several important quality control (QC) steps, like filtering low-quality cells based on the fraction of reads in peak regions (FRIP) score. We previously developed the method Chromap for ultrafast read alignment and preprocessing of scATAC-seq data. Chromap utilizes a sketch, which we call cache, to reuse alignment information for reads from peak regions to save running time. In this work, we augmented this cache structure to report important QC metrics, such as approximate FRIP score and cache occupancy. In particular, the FRIP score for a cell is predicted by a linear regression model considering the fraction of reads in the cache and other factors, like the number of duplicate reads in the cell. Cache occupancy is the number of distinct cache slots, inferred by the k-MinHash sketch, containing reads from the cell, which reflects the number of peaks for a cell. We show that these metrics capture additional low-quality cells missed by methods like SnapATAC2, and these metrics can be obtained without sacrificing running speed. The source code of Chromap is available at https://github.com/haowenz/chromap.

Keywords: Single-cell ATAC-seq, quality control, sketch

Weiman Yuan (Department of Biomedical Engineering, Vanderbilt University), Can Luo (Department of Biomedical Engineering, Vanderbilt University), Yunfei Hu (Department of Computer Science, Vanderbilt University), Liting Zhang (Florida State University), Zihang Wen (Carnegie Mellon University), Yichen Henry Liu (Department of Computer Science, Vanderbilt University), Xian Mallory (Department of Computer Science, Florida State University) and Xin Maizie Zhou (Vanderbilt University). *CNVeil enables haplotype-resolved copy number inference and clonal analysis in single-cell tumor genomes from single-cell DNA sequencing data.*

Abstract. Single-cell DNA sequencing (scDNA-seq) enables copy number variation (CNV) analysis at single-cell resolution, providing insights into tumor heterogeneity and subclonal structure. While most existing methods focus on total copy number, haplotype-resolved copy number is critical for characterizing allelic imbalance and clonal evolution. However, accurate inference of both total and haplotype copy number from scDNA-seq data remains challenging due to low coverage, allelic dropout, and technical variability.

We present CNVeil, a quantitative framework for joint inference of total and haplotype-resolved copy number from scDNA-seq data. CNVeil estimates total copy number by correcting systematic bias using normal cells identified via a PCA-based Gini coefficient, and performs robust ploidy estimation and segmentation through hierarchical clustering with cross-cell change rate–based breakpoint detection to reduce noise. Building on these segments, CNVeil infers allele-specific copy number by leveraging unphased single-nucleotide polymorphism counts under constraints imposed by total copy number within an expectation–maximization framework, thereby increasing statistical power. It further reconstructs chromosome-wide haplotype copy number profiles by phasing segments across the genome using a dynamic programming approach.

Across extensive benchmarking against 12 state-of-the-art methods on both simulated and real cancer datasets spanning diverse scDNA-seq technologies, including high-multiplexed single-nucleus sequencing, ACT, and 10x Chromium, CNVeil demonstrates strong performance in subclone identification, segmentation accuracy, and haplotype-resolved copy number inference. Together, CNVeil provides a unified framework for total and haplotype-resolved copy number analysis at single-cell resolution, enabling deeper characterization of tumor heterogeneity and clonal evolution.

Keywords: haplotype-resolved copy number, single-cell DNA sequencing, tumor heterogeneity, clonal evolution, expectation–maximization algorithm, cancer genomics

Arun Seetharam (Rosen Center for Advanced Computing, Purdue University, West Lafayette, IN 47906, USA), Laura Tibbs-Cortes (USDA-ARS, Corn Insects and Crop Genetics Research Unit, Ames, IA 50011, USA) and Margaret R. Woodhouse (USDA-ARS, Corn Insects and Crop Genetics Research Unit, Ames, IA 50011, USA). *HelixForge: Evidence-Based Reconciliation of Deep Learning Gene Predictions with RNA-seq and Protein Homology.*

Abstract. Genome annotation of large, repeat-rich plant genomes such as maize remains challenging. Traditional pipelines combining ab initio predictors, transcript evidence, and homology require species-specific training and extensive manual curation. Deep learning predictors like Helixer bypass training requirements but produce single-isoform predictions with uncharacterized failure modes, making them insufficient as standalone annotation sources. No existing tool systematically reconciles deep learning predictions with experimental evidence while preserving the predictor's strengths.

We present HelixForge, an open-source pipeline that produces publication-quality gene annotations by reconciling Helixer predictions with RNA-seq transcript assemblies and protein homology alignments. HelixForge implements a nine-step workflow: baseline scoring of every predicted locus before modification, PASA-inspired classification of transcript evidence overlaps, TSEBRA-inspired quality gating, pairwise competition where evidence-based modifications must outscore the original prediction to be accepted, a CDS assignment cascade that preserves predictor boundaries when possible, and score-gated splice site correction informed by the predictor's own per-base confidence signals. The pipeline supports parallel execution across HPC clusters via genome chunking.

We validated HelixForge using a six-phase error profiling analysis on Arabidopsis thaliana (Araport11 reference), where it improved exact CDS match rate from 73.8% to 76.9% (+874 genes), gene-level F1 score from 74.30 to 77.44, and intron chain F1 from 60.03 to 63.30, with a 5.6:1 improvement-to-regression ratio. Error profiling revealed that the predictor's minimum CDS confidence is the dominant error discriminator (AUC 0.862), directly informing HelixForge's confidence-based triage. We are now applying HelixForge to annotation of maize NAM founder genomes, where complex gene families, large introns, and paleopolyploidy present additional challenges beyond the Arabidopsis validation.

Keywords: Helixforge, gene annotation, gene model refinement, Helixer, evidence reconciliation

Shuyi Wang (THE CHINESE UNIVERSITY OF HONG KONG) and Yingying Wei (THE CHINESE UNIVERSITY OF HONG KONG). *Estimating a phylogenetic forest for single-cell RNA-sequencing data.*

Abstract. Single-cell RNA-sequencing (scRNA-seq) technologies provide researchers with unprecedented opportunities to identify cell types and understand cell lineages. With the emergence of scRNA-seq studies that assay a large number of subjects, there is growing interest in aligning and comparing cell lineages between different individuals, especially for those with different clinical conditions. However, comparing cell lineages learned from scRNA-seq data collected from multiple individuals is challenging due to (a) scRNA-seq data can suffer from severe batch effects and (b) certain cell types may occur in some but not all individuals. In this study, we propose a Bayesian hierarchical model built upon Dirichlet diffusion tree to learn a phylogenetic forest for scRNA-seq data collected from multiple individuals. Our proposed model can automatically correct batch effects caused by multi-batch measurements and align the topologies of the phylogenetic trees of different individuals. We develop an efficient Markov chain Monte Carlo algorithm for posterior inference. Real data analysis shows that the proposed method can effectively reconstruct and compare cell lineages across samples. Thus, we envision that the proposed method will greatly facilitate the study of development, cellular heterogeneity, and disease-related changes with population-scale scRNA-seq studies.

Keywords: Single-cell RNA-sequencing data, Phylogenetic tree, Dirichlet diffusion tree, Bayesian hierarchical model

Anil Prakash (Virginia Tech) and Shenglin Mei (Virginia Tech). *Localized Pseudobulk Analysis of Single-Cell Multi-Omic Divergence via Pi-Score Integration.*

Abstract. The emergence of single-cell multi-omic technologies has provided unprecedented views into the regulatory mechanisms of disease. However, identifying robust biological divergence at the single-cell level remains a statistical challenge due to technical noise and the trap of pseudoreplication. We present a localized pseudobulk framework designed to extract high-confidence biological signals from integrated single-cell RNA and ATAC-seq data.

Our method leverages a localized neighborhood aggregation strategy that transforms sparse single-cell profiles into batch-aware pseudobulks. By performing differential analysis within these specific cellular niches and applying empirical Bayes variance moderation, we stabilize the variance across distinct biological replicates. A central feature of this model is the calculation of a signed Pi-score—a composite metric that integrates both the magnitude of fold-change and statistical significance. This score allows for the ranking of gene expression and chromatin accessibility changes simultaneously, effectively highlighting the most biologically relevant features that drive cellular divergence.

When applied to multi-omic datasets, this approach effectively identifies Differentially Accessible Peaks (DAPs) and Differentially Expressed Genes (DEGs) that are consistent across batches. The resulting cell-divergence scores provide a quantitative landscape of how specific cell types deviate from healthy baselines in response to disease or stimuli. By prioritizing the Pi-score, our model transcends simple p-value thresholds, offering a more nuanced interpretation of the regulatory landscape. This framework provides a significant tool for researchers to pinpoint critical regulatory nodes in complex, multi-modal single-cell atlas data.

Keywords: single-cell, multi-omic, localized differential analysis, pi-score, regulatory

Zhiang Yuan (The Chinese University of Hong Kong), Hongyuan Cao (Florida State University) and Yingying Wei (The Chinese University of Hong Kong). *A Robust and Computationally Scalable Method for Integrating a Large Number of Genomic Studies.*

Abstract. While individual genomic studies can suffer from weak signals, genomic studies spanning diverse conditions are becoming increasingly prevalent. For example, the integration of multi-ethnic genome-wide association studies is now essential to bolster power in underrepresented minority cohorts. Meanwhile, technologies like Perturb-seq have enabled genome-scale screens of transcriptional responses across hundreds of knockouts. However, conducting meta-analysis across numerous conditions presents significant challenges: the total number of association patterns 2^D grows exponentially with the condition number D. Moreover, existing meta-analysis methods typically rely on parametric assumptions that are not applicable to Perturb-seq studies, which usually adopt nonparametric tests for differential expression. Here, we propose a novel method that is able to (a) detect condition-specific signals while borrowing strength across conditions, (b) scale to a large number of conditions and (c) estimate non-null p-value distributions nonparametrically. We establish model identifiability and asymptotic false discovery rate control. Numerical studies show that our method outperforms existing methods.

Keywords: Meta-analysis, Gene expression, Genome-wide association studies, Perturb-seq, False discovery rate, EM algorithm

Daniel Hupalo (Uniformed Services University, Bethesda, MD), Jacob McCauley (Miller School of Medicine, University of Miami), Matthew Wilkerson (Uniformed Services University, Bethesda, MD), Clifton Dalgard (Uniformed Services University, Bethesda, MD) and William Scott (Miller School of Medicine, University of Miami). *The NeuroBioBank Whole-Genome Catalog : sequencing from human brain donors with nervous system disorders.*

Abstract. Nervous system disease is a prevailing cause of morbidity and death worldwide. This set of neurological and psychological diseases have diverse causes aided by unknown genetic risks. Here we generated genome-wide genetic data on a large cohort of brain tissue donors with well annotated medical histories, allowing for broad investigations into the genetic risk of developing these neurological, neurodevelopmental, and psychological conditions. This resource consists of 9,663 donors with whole-genome genotyping and 9,543 donors with whole-genome sequencing (WGS) completed. The clinical diagnoses of these donors include 148 nervous system diseases clustered in 15 broad categories by ICD-10 coding. These donors were collected by six repositories comprising the NIH NeuroBioBank, with an average participant age of 60 years old. While primarily older individuals of European descent, the cohort also contains younger donors and individuals from non-European backgrounds. Full demographic and phenotypic descriptions for all donors are available via the NeuroBioBank Portal (neurobiobank.nih.gov). Variants detected by WGS were jointly called genome-wide and annotated to describe their functional impact; this identified 1,047,815 non-silent mutations. To illustrate applications of this resource, we replicated the strong associations observed in previous studies for single variants, structural variants, and burden tests for diseases such as Huntington's disease and Alzheimer's disease.

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Keywords: Whole-genome sequencing, brain, psychiatric disease, neurological disease, WGS Cohort

Christie Rose Woodside (The George Washington University). *Quality Control Metrics for Selecting Reference-grade Genomic Data for Medical Countermeasures and Diagnostic Use Cases.*

Abstract. The rapid growth of genomic data in public repositories has enabled advances in infectious disease diagnostics and biosurveillance, but variability in genome assembly quality and inconsistent metadata limit the selection of reliable reference datasets. Current quality control (QC) approaches remain fragmented and lack standardized criteria for regulatory or clinical use.

We present a scalable computational framework developed within the FDA-ARGOS initiative to standardize QC evaluation of genomic data. This framework integrates three complementary pipelines, ngsQC, assemblyQC, and biosampleMeta, within a reproducible workflow documented using BioCompute Objects to ensure provenance and FAIR-compliant data practices. A versioned data dictionary comprising 272 QC attributes was applied to 1,430 genome assemblies across 524 species from our NCBI BioProject and curated pathogen datasets.

To evaluate redundancy and identify informative QC features, we performed principal component analysis with varimax rotation on both read-level and assembly-level metrics. Results show that a reduced subset of orthogonal QC metrics captures more than 85 percent of the variance in genome quality, enabling dimensionality reduction while preserving interpretability. Assembly-level metrics exhibited greater independence compared to read-level metrics, suggesting distinct contributions to overall quality assessment.

This work provides a standardized and reproducible approach for evaluating genomic data quality and selecting reference-grade assemblies for diagnostic and regulatory applications. The resulting QC-annotated datasets are publicly available and enable researchers to identify and use high-quality reference genomes for biosurveillance, assay development, and next-generation sequencing diagnostics.

Keywords: genomic quality control, reference genome curation, high-throughput sequencing, biosurveillance, diagnostic genomics, reproducible workflows, BioCompute Objects, FAIR data, principal component analysis, genome assembly quality

Minh Le (The University of Texas MD Anderson Cancer Center School of Health Professions).

Population-Structure-Aware SNP Threshold Derivation Improves Outbreak Resolution in Escherichia coli.

Abstract. Pairwise single nucleotide polymorphism (SNP) distances derived from whole genome sequencing (WGS) data are the quantitative measure for inferring potential pathogen transmission. However, SNP thresholds are often applied arbitrarily across bacterial species, causing false positive or false negative transmission calls. With the increasing accuracy of Oxford Nanopore Technologies (ONT), we sought to develop an ONT-only approach to derive population structure aware SNP thresholds for Escherichia coli subtypes.

We evaluated long-read variant calling on simulated data and found Clair3 with a bacteria- finetuned model showed the highest SNP calling accuracy. For benchmarking, we sequenced 12 index–recurrent pairs using Illumina and ONT. Clair3 long-read calls were highly concordant with Snippy short-read calls, and long reads resolved SNPs in repetitive regions where short reads could not map uniquely.

We applied the ONT only workflow to 500 clinical isolates, including 180 index–recurrent pairs as a surrogate for "transmitted strains." Core genome MLST allele distances were used for hierarchical clustering and split decomposition to identify genomic substructure without imposing a fixed tree. We delineated subgroups within ST131, a highly prevalent and clinically relevant lineage. Within each subgroup, recombination masked SNP distances showed clear separation between within patient and between patient comparisons. SNP thresholds at the 75th or 90th percentile yielded higher sensitivity and specificity within subgroups compared to thresholds applied to ST131 as a whole.

These results show that ONT only WGS can accurately quantify SNP distances and that population structure aware thresholds improve ST131 subgroup-specific transmission inference, suggesting clustering-based approaches may improve genetic relatedness measurements across diverging bacterial lineages.

Keywords: long reads, variant calling, genetic relatedness, wgs-inferred outbreak transmission analysis, Escherichia coli

Yanmei Dou (Westlake University). *Detection of Somatic Point Mutations Directly from Spatial Transcriptomics Enables in vivo Spatiotemporal Lineage Tracing.*

Abstract. Spatial transcriptomics reveals tissue organization but lacks in vivo lineage-tracing methods applicable to humans. We introduce SpaceTracer, a computational framework that accurately detects somatic single-nucleotide variants (SNVs) directly from spatial transcriptomics data. By leveraging naturally occurring somatic SNVs, SpaceTracer reconstructs cellular phylogenies within native tissue architecture, enabling the mapping of lineage spread, migration, lineage-coupled expression changes and lineage-aware local interactions. Applied to human cutaneous squamous cell carcinoma, it traced tumor initiation and progressions, uncovered widespread pre-invasive migration of dedifferentiated epithelial cells and characterized mutant B cells migrating from tertiary lymphoid structures (TLS) into the tumor boundary. The framework also reconstructed developmental lineages across multiple tissues and identified tissue-resident mutant immune cells. SpaceTracer thus provides a perturbation-free platform for high-resolution spatiotemporal lineage tracing, offering a transformative tool for elucidating complex biological systems—especially tumor-immune ecosystems—with direct implications for advancing cancer immunotherapy.

Keywords: spatial transcriptomics, somatic mutation, lineage tracing

Cherie Tsz Yiu Law (School of Biomedical Sciences, Faculty of Medicine, The Chinese University of Hong Kong, Hong Kong) and Stephen Kwok Wing Tsui (School of Biomedical Sciences, Faculty of Medicine, The Chinese University of Hong Kong, Hong Kong). *Integrated host-pathogen multi-omics reveals mechanisms underlying fatal Balamuthia mandrillaris encephalitis.*

Abstract. Balamuthia mandrillaris is a free-living amoeba that causes highly progressive fatal encephalitis, yet how this environmental protist invades and adapts within the human brain remains poorly understood. Here, we integrate a de novo genome from clinical brain lesions with in situ host-parasite multi-omics profiling to identify molecular features associated with invasion, immune evasion, and metabolic adaptation of the amoebae. We identify an expanded set of secreted M12B disintegrin metallopeptidases enriched in the patient brain lesion that mediates traversal across the extracellular matrix, and a secreted calreticulin homolog that engages complement component C1q. In parallel, single-nucleus profiling of the patient's brain lesion reveals broad metabolic suppression across neuronal, glial, and vascular compartments. Macrophages are shifted toward an immunoregulatory $APOE^{+}$ phenotype rather than an amoebicidal pro-inflammatory state. Under metabolic stress, B. mandrillaris relies on the alternative oxidase pathway, where specific inhibition disrupts the formation of natural cyst resistance (drug-resistant cysts), exposing a therapeutic vulnerability. Collectively, these results suggest that environmental survival traits of B. mandrillaris can contribute to fatal encephalitis and provide a framework for dissecting rare and neglected infections.

Keywords: Balamuthia amoebic encephalitis, free-living amoeba, multi-omics, infectious disease

Liangliang Gao (Plants for Human Health Institute, North Carolina State University, USA), Romit Seth (Plants for Human Health Institute, North Carolina State University, USA), Kopalli Venkataramana (Department of Agrobioinformatics, Liebig University, Germany), Agnieszka Golicz (Wageningen University & Research, Netherlands), Dariusz Grzebelus (Department of Plant Biology and Biotechnology, University of Agriculture in Krakow, Krakow, Poland), Phillip Simon (Dept of Plant and Agroecosystem Sciences, UW-Wisconsin and USDA-ARS, Vegetable Crops Research Unit, Madison, WI, USA) and Massimo Iorizzo (Plants for Human Health Institute & Department of Horticultural Science, North Carolina State University, USA). *Structural Variant Mapping Enhances Genetic Discoveries for Pigmentation in Daucus carota.*

Abstract. While single nucleotide polymorphisms (SNPs) are the standard markers for trait mapping, they often fail to capture the linkage with important traits. Recent work in plants highlighted the value of Structural Variants (SVs) to address this challenge. Despite their significance, no study has addressed the genome-wide effects of SVs on GWAS in carrot. This study integrates a large resequencing dataset to resolve the genetic basis of root carotenoids and anthocyanin using SVs. Based on a single reference genome, over 100,000 high-quality SVs were identified across a diverse panel of 630 carrot accessions. Comparative SV-GWAS and SNP-GWAS reveal that structural variants capture significant "missing associations" for alpha carotenoid content and purple root pigmentation that were not detected by point mutations. Closer inspection of the region spanning the SVs association with purple root pigmentation identified a well-known anthocyanin related transcription factor, which support the effectiveness of the SVs association. These results demonstrate that SVs provide a critical, complementary strategy to uncover the genetic mechanisms underlying complex traits. Ongoing work will leverage a wider set of germplasm and a pangenome graph-based SVs to expand and refine the dissection of metabolite traits in carrots.

Keywords: Structural Variants, Single Nucleotide Polymorphisms (SNPs), Carrots, Anthocyanin, Carotenoid

Jason Miller (Hood College), Karina Hornslien (University of Oslo) and Paul Grini (University of Oslo).

KIRP: Detecting allele-specific expression with k-mer analysis of RNAseq data.

Abstract. Epigenetic effects such as genomic imprinting have been shown to play a role in the endosperm of developing seeds of the model organism, Arabidopsis thaliana. Experiments on other plant species must overcome challenges of outcrossing, heterozygosity, and lower quality references. To expand the study of allelic expression to other Arabidopsis species, we present a new protocol that has low reliance on reference data. KIRP uses k-mer analysis of paired RNAseq read sequences from three individuals: two parents and their cross. It builds k-mer sets specific to each parent and deletes their intersection. It assigns cross reads to a parent, where possible, by set comparison. It bins reads by gene using a map to the nearest available annotated reference. It applies standard differential expression statistics to the binned read counts. KIRP was designed to avoid the pitfalls of map bias that arise when parental genomes differ by level of heterozygosity and similarity to the (single-haplotype) reference. We present KIRP results on crosses of two wild ecotypes of the species Arabidopsis arenosa. We replicated experiments on four crosses in each direction (i.e. swapping the pollen donor and acceptor), and we filtered for genes expressed in endosperm. Compared to a reference-based analysis of the same data, the KIRP results indicated greater resilience to the confounding factors of heterozygosity and map bias.

Keywords: Allele-specific expression, Arabidopsis, genomic imprinting, RNAseq, k-mer analysis

Yu-Nong Gong (Chang Gung University). *From Raw Sequences to Taxonomic Inference: An End-to-End Hybrid Framework for Uncovering Cryptic Viral Elements.*

Abstract. Identifying uncharacterized viruses in high-throughput sequencing datasets remains a significant computational bottleneck. Traditional homology-based tools depend heavily on reference genomes, often failing to detect cryptic viral elements that lack significant sequence similarity to known pathogens. Furthermore, conventional pipelines yield complex tabular data that require extensive manual curation. To address this, we developed an automated Snakemake pipeline designed to maximize viral rescue through a hybrid diagnostic architecture. The framework integrates: (1) geNomad for neural network-driven pattern recognition; (2) HMMscan for identifying conserved protein domains in Pfam; and (3) CheckV for genome completeness estimation and host contamination filtering. To streamline analysis, a Large Language Model (LLM) module leverages these quantitative metrics to derive evidence-based conclusions, linking detected biological features to taxonomic assignments. We benchmarked the pipeline using a mock dataset of 2,500 sequences, including 500 uncultivated viral and 2,000 non-viral entries. This simulates the challenge of discovering highly divergent or novel zoonotic pathogens, such as undiscovered members of the Coronaviridae family. While standalone tools demonstrate baseline sensitivities within the 90.4%–96.0% range, this hybrid framework yielded a 98.8% recovery rate, successfully rescuing divergent sequences that would otherwise be lost. By integrating compositional, functional, and genomic signatures into a reproducible workflow, this framework provides the high sensitivity required for uncovering cryptic viral elements in complex metagenomic datasets where reference homology is limited.

Keywords: Metagenomics, High-throughput Sequencing, Pathogen Discovery, Large Language Models, Sequence Homology

Xuecong Fu (Foundation Medicine Inc), Yu Sun (Foundation Medicine Inc.), Neil Peterman (Foundation Medicine Inc.) and Alex Robertson (Foundation Medicine Inc.). *Accurate methylation calling from damaged DNA.*

Abstract. There are many applications of analyzing DNA methylation patterns in cell free DNA (cfDNA) in patients with cancer. They can inform on the epigenetic state of the primary tumor and its metastases, giving insight into how well the patient is responding to therapy and which therapies the tumor may be susceptible to. Methylation patterns are assayed by subjecting the cfDNA to a conversion process that converts each unmethylated cytosine into uracil which is read as thymine. Existing algorithms and code packages exist to analyze the raw data from such experiments. The majority of cfDNA molecules have characteristic jagged ends (double-stranded DNA molecules with protruding single-stranded ends) and accumulate DNA damage marks such as nicks (single-strand breaks in the DNA backbone) in circulation. However, library preparation methods often seek to fill in jagged ends or replace nucleotides downstream of nicks by incorporating unmethylated cytosines into the molecule. Such practice incorporates artifactual unmethylated cytosines in the template, resulting in "methylation bias" which confounds detection of true methylation patterns. We present computational methods to identify methylation signals in Watson/Crick strands of input DNA molecules likely introduced by jagged ends/nick repair or insufficient conversion in library preparation. By excluding these confounding regions in the analysis, we reconstruct a dependable portion of the original cfDNA molecule for downstream methylation calls. We also present data showing that these estimates are concordant with an orthogonal measure of cfDNA damage. Taken together, these methods could potentially improve the accuracy of methylation analysis in cfDNA.

Keywords: DNA methylation, cell free DNA, cancer detection

Francesco Nicassio (Istituto Italiano di Tecnologia). *Integrative lineage tracing, long-read transcriptomics and 3D epigenomics reveal functional regulators of TNBC plasticity.*

Abstract. Plasticity is a defining hallmark of triple-negative breast cancer (TNBC), enabling tumors cells to shift between proliferative and invasive states, acquire stem-like traits, and escape treatment. We previously showed, using single-cell multi-omic lineage tracing integrated with phenotypic assays, that aggressive phenotypes including tumors-initiating potential and drug tolerance can be pre-encoded within rare clonal subpopulations in TNBC models. However, the regulatory circuits linking clonal fate, chromatin state, and transcriptional output remain incompletely understood.

Here, we present an integrative framework to dissect the regulatory architecture of TNBC plasticity by combining 2D and 3D epigenomic profiling, long-read transcriptomics, and CRISPR-based perturbation. Building on tumors-initiating clones identified by lineage tracing, we profiled the epigenetic features underlying this pre-adapted state relative to the bulk tumor population. Histone-mark profiling and H3K27ac HiChIP identified regulatory regions and enhancer-promoter contacts associated with the tumor-initiating program, consistent with specific 3D genome reorganization in these clones. RADICL-seq provided orthogonal support by detecting proximal and distal RNA-chromatin interactions at candidate loci.

These maps are complemented by a long-read transcriptomic and epitranscriptomic resource for breast cancer that enables high-confidence transcript annotation and resolves isoform-specific programs, previously unannotated lncRNAs, and enhancer-associated transcripts in aggressive TNBC states. CRISPR-based perturbation then enabled causal testing of candidate dependencies, highlighting actionable non-coding regulators associated with hEMT-like plasticity and tumor-initiating potential. Overall, this work defines a multilayered framework linking clonal fate, chromatin rewiring, and functional dependencies in cancer plasticity, providing a general strategy to decode how pre-adapted cancer states emerge from integrated epigenome and transcriptome regulation.

Keywords: cancer plasticity, lineage tracing, 3D epigenomics, enhancer-promoter interactions, long-read transcriptomics, CRISPR perturbation, non-coding regulation, gene expression

Raighen Ly (Hope College), Robert Glowacki (Hope College) and Catherine Welsh (Hope College).

Resolving Novel Bacteroides thetaiotaomicron Genomes Using Hybrid Assembly and Assembly Graph Analysis.

Abstract. Bacteroides thetaiotaomicron (B. theta) is widely used as a model organism for studying host-microbiome interactions, yet most genomic analyses rely on the reference strain Bt-VPI-5482 despite substantial strain-level variation. Two additional strains, Bt-5951 and Bt-0940-1, currently lack publicly available closed genome assemblies, limiting comparative analyses and interpretation of strain-specific phenotypes.

These strains along with Bt-3164 and Bt-VPI-5482 were sequenced and their genomes were assembled using a hybrid assembly workflow combining short- and long-read sequencing with Unicycler. Short reads were quality filtered and resulting assembly graphs were analyzed to identify unresolved repeats and ambiguous paths preventing chromosome closure. Assembly quality was evaluated using metrics from Quast and visually inspected with Bandage.

Our current assemblies have reduced each strain to three to five contigs and assembly graph analysis suggests that the remaining breaks are associated with repeated regions and plasmids. Ongoing finishing is expected to produce complete circular chromosomes prior to presentation of this work. Preliminary comparisons to Bt-VPI-5482 indicate both conserved genomic structure and candidate strain-specific differences. The resulting assemblies will provide new genomic resources for studying strain-level diversity in B. theta and illustrate practical strategies for closing challenging bacterial genomes.

Keywords: Bacterial genome assembly, Hybrid genome assembly, assembly graph, Bacteroides thetaiotaomicron, gut microbiome

Atieh Abedin-Do (Cedars Sinai Medical Center), Linbu Liao (Shenzhen University of Advanced Technology), Xing Ye (Cedars Sinai Medical Center) and Xue Li (Cedars Sinai Medical Center). *A multiomic and spatial atlas of mouse cloaca development reveals an early Gata6-driven regulatory program governing bladder mesenchymal specification.*

Abstract. How the cloaca gives rise to the bladder and genital tubercle remains poorly understood, largely because cell-type–specific regulatory programs across developmental time have not been resolved. We generated a multiomic and spatially resolved atlas of mouse cloaca development spanning embryonic days E10.5–E15.5 by integrating single-nucleus RNA/ATAC sequencing , and Visium HD spatial transcriptomics.

Across 113,989 cells, we identified epithelial and mesenchymal populations and reconstructed their developmental trajectories. Joint RNA/ATAC embedding enabled robust cell-type annotation and revealed stage-specific regulatory landscapes. We identified a bladder-directed mesenchymal population characterized by strong enrichment of Gata family motifs (Gata6/5) and coordinated transcriptional activation.

Integrative analysis of chromatin accessibility, gene expression, and peak-to-gene associations prioritizes Gata6 and Gata5 as central regulators of bladder mesenchymal specification. This regulatory program is supported across modalities: motif accessibility, transcriptional expression, and enrichment of known Gata6 target genes.

In contrast, genital tubercle mesenchyme follows a distinct regulatory trajectory during early development (E10.5–E12.5), marked by transient Lef1 activity and subsequent engagement of alternative transcriptional programs, such as Dlx5/3.

Spatial transcriptomic mapping independently confirms that predicted bladder and genital tubercle populations localize to anatomically distinct regions, providing spatial validation of single-cell–derived identities in tissue context.

This work establishes a stage-resolved multiomic framework for cloaca development and identifies a Gata6-driven regulatory program governing early bladder mesenchymal specification; one that has not previously been characterized at this resolution.

Keywords: Spatial transcriptomics, single-nucleus multiomic sequencing, Bladder, Genital tubercle, Development

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Pablo Atienza (Genomics of Gene Expression Lab) and Ana Conesa (Institute for Integrative Systems Biology, Spanish National Research Council). *Guiding structural gene prediction with long-read transcriptomics using SQANTI Evidence.*

Abstract. Despite the rapid advancement of long-read transcriptomics, current genome annotation methods often struggle to fully capture the structural complexity of the transcriptome. While long-read sequencing provides high resolution of full-length isoforms and alternative splicing events, integrating these data into traditional annotation workflows remains a significant bottleneck. Existing pipelines frequently rely on short-read data or generalized ab initio models, which can result in fragmented gene structures and the mischaracterization of novel transcripts. To address this limitation, we present SQANTI Evidence: a novel, evidence-driven genome annotation pipeline that directly integrates long-read sequencing data to accurately guide structural gene prediction. By incorporating transcript-level features to validate and refine gene models, our tool produces high-fidelity, comprehensive reference annotations.

Our approach builds on the findings of Paniagua et al. (2025), where we demonstrated that using long reads as a guiding scaffold, rather than the sole source of evidence, yields superior results. SQANTI Evidence has been implemented using Snakemake (Mölder et al., 2021) as the workflow control tool. Built upon the established framework of AUGUSTUS (Stanke et al., 2008), the pipeline uses an IsoSeq3 transcriptome derived from long-read sequencing, curated using the SQANTI suite, as external evidence to guide the gene discovery process. This integration enables the generation of annotations that accurately reflect alternative splicing and isoform diversity. In addition to structural gene annotation, SQANTI Evidence produces a comprehensive quality report using GAQET2, a novel quality control pipeline that allows the user to obtain multiple parameters for a single annotation and benchmark against currently available state-of-the-art methods.

SQANTI Evidence has been benchmarked against a set of model eukaryotic organisms, including human, mouse, fish, and yeast. The data for each annotation were taken from public datasets, and the reference genomes from ENSEMBL. In human datasets, perhaps the most well-studied of the organisms tested, the annotation benchmark achieved higher levels of sensitivity and structural resolution than standard annotation tools like BRAKER3 (Gabriel et al., 2024), which was run on the recommended mode for long-read RNA data. It should be mentioned, that this “mode” requires of proteomics data to be run, thus using a slightly different input from SQANTI Evidence, which solely relies on transcriptomic input and the reference genome.

Using the whole human genome as a benchmark due to its massive scale and high transcriptomic complexity, SQANTI Evidence showed a substantial increase in sensitivity and overall F1 score at the exon level, as can be seen in Table 1. While BRAKER3 achieved an exon-level sensitivity of 39.2% and a precision of 89.3% (F1 score: 54.5%), SQANTI Evidence boosted sensitivity to 52.0% and achieved a superior F1 score of 60.9%, despite a trade-off in precision (73.5%). Furthermore, the direct integration of long-read transcriptomic evidence substantially reduced the number of false negatives. BRAKER3 failed to capture 40.2% (123,206) of the known reference exons; in contrast, SQANTI Evidence nearly halved this error rate, missing only 24.3% (74,516). These results highlight the capacity of SQANTI Evidence to resolve complex splicing events and capture a more complete picture of the transcriptome compared to traditional methods.

SQANTI Evidence provides a robust framework for generating high-quality genome annotations that capture the full complexity of transcriptomic data, establishing accurate gene models to support downstream functional genomics across a wide range of species. Future benchmarking will be expanded to include non-model organisms and more annotation tools, especially in matters of speed and resource usage.

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Keywords: Gene annotation, Long reads, Structural annotation

Flywell Kawonga (Malawi- Liverpool Wellcome Research Programme). *The first whole-Genome Sequences of human Sapovirus from Blantyre, Malawi reveal co-circulating Genogroups and Genotype Diversity.*

Abstract. Background

Human sapoviruses (SaVs) are increasingly recognised causes of paediatric acute gastroenteritis (AGE) worldwide, yet remain genomically under-characterised in Africa. In Malawi, limited whole-genome data restrict our understanding of SaV genetic diversity and evolution.

Materials and Methods

Paediatric AGE stool samples from Blantyre were screened using customised real-time PCR TaqMan Array Cards (TACs). SaV-positive samples (Ct<35) underwent RNA extraction, cDNA synthesis, shotgun metagenomic library preparation, and Illumina NextSeq 2000 sequencing (2×150 bp). Reads were trimmed, human reads removed, and sapovirus genomes assembled and confirmed by annotation, genotyping and phylogenetics.

Results

SaV was detected in 6.23% (60/963) of AGE cases, with diverse viral co-detections. We generated the first SaV whole-genome sequences from Blantyre, Malawi (n=14), identifying co-circulation of GI, GII and GIV. GII.5 was most frequent (28.6%, 4/14), followed by GI.1 (21.4%, 3/14), GI.2 (14.3%, 2/14) and GII.1 (14.3%, 2/14); GIV.1, GII.2 and GI.6 were each detected once.

Conclusions

Sapovirus was identified among paediatric AGE cases in Blantyre and showed genomic diversity, with multiple co-circulating genogroups and genotypes. These first whole-genome data from Blantyre, Malawi provide a foundation for sapovirus genomic surveillance in Malawi and across Africa, enabling studies of transmission, evolution and recombination, and supporting improved diagnostics and future vaccine development.

Keywords: acute gastroenteritis, Africa, genomic surveillance, Malawi, sapovirus

Kumar Arvind (Division of Intramural Research, National Library of Medicine, Bethesda, MD, USA), Laura Heinly (Human Brain Collection Core, NIMH-IRP, Bethesda, MD, USA), Ningping Feng (Human Brain Collection Core, NIMH-IRP, Bethesda, MD, USA), Anton Schulmann (Human Brain Collection Core, NIMH-IRP, Bethesda, MD, USA), Ajeet Mandal (Human Brain Collection Core, NIMH-IRP, Bethesda, MD, USA), Pavan Auluck (Human Brain Collection Core, NIMH-IRP, Bethesda, MD, USA), Stefano Marenco (Human Brain Collection Core, NIMH-IRP, Bethesda, MD, USA) and Yun Zhang (Division of Intramural Research, National Library of Medicine, Bethesda, MD, USA). *A Spatially Informed Cellular Reference Map of the Human Amygdala.*

Abstract. The human amygdala is a complex brain structure composed of approximately 13 anatomically distinct subnuclei, the major subnuclei being the central (CeA), basal (BA), and lateral (LA) nuclei. It is responsible for behavioral responses to emotional stimuli. Alterations in amygdala have been linked to several neurodegenerative and psychiatric disorders. Understanding the cellular organization of human amygdala is crucial to elucidating its role in disease states. To this end, we leveraged single-nucleus RNA sequencing (snRNA-seq) and spatial transcriptomic approaches to generate a spatially-resolved cell atlas of the human amygdala.

Transcriptional profiles of approximately 30,000 spatially indexed nuclei were analyzed using standard snRNA-seq workflow. Broad cell classes identified were excitatory neurons, inhibitory neurons, astrocytes, oligodendrocytes, oligodendrocyte precursor cells, microglia, and endothelial cells. Molecularly defined broad cell classes were visualized on the tissue sections using their respective Trekker spatial coordinates. Spatial projection revealed that neuronal and non-neuronal cell types were distributed uniformly at the broad cell class level across the tissue map. Notably, regions enriched in inhibitory neuron were identified on the spatial map, suggesting the potential location of Central Amygdala (CeA).

Further analyses will characterize neuronal subtypes at granular level leveraging the NS-Forest machine learning framework. Future integration of PIP-seq (snRNA-seq) and Visium (spatial transcriptomics) datasets of the same region will be conducted to enable fine-grained spatial mapping of amygdala cell populations. Together, this study anticipates establishing a spatially informed atlas of human amygdala centered on CeA that would serve as a reference for localizing cell type-specific transcriptional changes in spatial structure.

Keywords: Brain, Amygdala, Single-nucleus transcriptomics, Spatial transcriptomics, Spatially informed cell atlas

Zhenhua Shang (University of Louisville), Jae Yeon Hwang (University of Louisville), Ming Song (University of Louisville) and Juw Won Park (University of Louisville). *Exploring the Spatial Transcriptome at Subcellular Resolution via Visium HD.*

Abstract. Background

Spatial transcriptomics preserves gene expression within intact tissues, revealing organization lost in traditional sequencing. While bulk RNA-seq averages signals and single-cell RNA-seq (scRNA-seq) requires tissue dissociation, both sacrifice spatial context. Visium HD addresses these gaps through unbiased, whole-transcriptome profiling at subcellular resolution, enabling precise mapping of cellular niches and microenvironmental dynamics.

Methods

Fresh-frozen mouse small intestine sections were processed using the Visium CytAssist workflow. Data were binned at 8 µm to balance sensitivity and resolution. To mitigate challenges like sparse transcript counts and partial volume effects, we employed a biology-driven annotation strategy. Cell types were manually identified using canonical markers validated by immunohistochemistry (IHC), in situ hybridization (ISH), and histological cross-referencing, ensuring more robust assignments than purely computational deconvolution.

Results

The analysis successfully resolved the crypt-villus architecture, identifying ten major cell populations with precise localization. Clustering revealed spatial domains that align with histological layers, capturing fine-scale signaling gradients and immune-stromal interactions. These insights represent biological details typically lost through bulk averaging (RNA-seq) or scRNA-seq dissociation.

Statement of Significance

This study demonstrates that combining 2 µm feature capture with rigorous manual annotation mitigates artifacts in high-resolution data. By bridging the gap between transcriptomic density and biological validation, this framework provides a high-fidelity map of cellular niches. This methodology offers a reliable approach for investigating tissue organization and spatial reprogramming in complex biological systems.

Keywords: spatial transcriptomics, Visium HD, Cellular niches

Aditya Aysola. *Binary Search as a Foundational Primitive in Genomic Sequence Alignment: A Computational Analysis.*

Abstract. The binary search algorithm is a foundational algorithm in computer science. It enables search over a sorted dataset in a worst-case O(log n) time. While its application to integer arrays is commonly taught and understood, its applications in biological sequence alignment are less commonly appreciated. This paper will examine how binary search underlies genomic alignment, tracing algorithmic progression from naive string matching to suffix arrays to the Burrows-Wheeler Transform. We demonstrate that the central challenge of genomic alignment is fundamentally a problem of imposing sorted order on biological sequence data, such that binary search can be applied efficiently at the genome scale. Through complexity analysis, we show that this progression reduces the alignment time from O(nm) naive string matching to O(m log n) with suffix arrays, enabling the alignment of hundreds of millions of short reads against a 3.2 billion base-pair reference genome in practical time.

Keywords: Binary, Alignment, Genomic, Sequence, Algorithmic, Biotech, Healthtech

Nathaniel Brown (Johns Hopkins University) and Benjamin Langmead (Johns Hopkins University). *Orbit: Bitpacked Move Structure Library for Efficient Pangenome Indexes.*

Abstract. Compressed indexes are increasingly used to answer pattern-matching queries within a memory budget, such as in pangenomics where users commonly need to search for substrings within large collections of related genome sequences. Even when a pangenome is on the level of terabytes, its Burrows-Wheeler Transform (BWT) can often be compressed to be much smaller. Examples which use the run length-encoded BWT (RLBWT) for their indexes include classification of DNA reads, sequence alignment, and haplotype similarity.

Recently, the move structure has advanced both theory and practice for RLBWT based indexes, thanks to its cache efficient locality of reference. However, not all tools have transitioned to using a move structure, and those that do have been custom built for their use cases, even though the underlying structure is similar. Further, the "splitting" required for fast and small move structures with theoretical guarantees may invalidate the data users need to store alongside to support pattern matching. To this end, we introduce the Orbit C++ library, a header only plug and play implementation of move structures.

Due to bitpacking, Orbit's index components are small, even when introducing user data types. The library also manages splitting regimes for ease of use. Further, we introduce length capping, a splitting optimization resulting in smaller and faster move structures in theory and practice. Evaluating on chromosome19 haplotypes, we find that length capping with Orbit results in faster and smaller data structures, sometimes much faster for certain queries, and at least 40% smaller on pangenomic scaled data.

Keywords: Compressed indexing, Pangenomics, Sequence alignment

Gamze Bulut (Virginia Commonwealth University), Isabella Napoli (Virginia Commonwealth University), Lavanya Sharma (Virginia Commonwealth University) and Johana Johnson (Virginia Commonwealth University). *Smooth Muscle Klf4 Promotes Spp1^{+} Macrophage Accumulation in Obese Adipose Tissue.*

Abstract. Background: Obesity remodels adipose tissue macrophages (ATMs), but how they respond to signals from neighboring smooth muscle cells (SMCs) is less clear. Here we re-analyzed multiple single-cell RNA-seq experiments to build a unified view of adipose immune and stromal cells in a mouse model with SMC-specific Klf4 deletion.

Methods: We combined 16 10x Genomics libraries from three experiments, including chow and diet-induced obesity (DIO) as well as wild-type (WT) and smooth-muscle Klf4 knockout (Klf4-SMC-KO) mice on DIO. After quality control, 27,398 cells remained. Datasets were integrated with SCTransform and canonical correlation analysis (Seurat v5), using experiment as a batch variable. We then subsetted 12,045 macrophage and mural cells based on marker gene expression and reclustered them using 12 principal components. We curated markers from the adipose macrophage literature (Itgax/CD11c, Cd9, Cd63, Lpl, Steap4, Mafb, Cybb, Pdgfra, Jam2, Prox1, Myocd) and used those to annotate clusters and assess diet- and genotype-dependent changes.

Results: Single-cell RNA-seq of adipose SVF revealed 18 transcriptionally distinct clusters including macrophages and mural cells. Cluster 0, the largest cluster, contained an Spp1-high macrophage population previously linked to pro-inflammatory and pro-fibrotic functions, crown-like structures, and insulin resistance. In our dataset, Spp1^{+} macrophages increased in DIO vs chow and were more abundant in Klf4 WT than in Klf4-SMC-KO mice, suggesting that SMC Klf4 may promote accumulation of this potentially maladaptive macrophage population, which has also been implicated in vascular dysfunction.

Conclusion: This integrated analysis, built collaboratively with undergraduate students, defines a macrophage- and mural-cell–focused atlas of adipose stromal vascular cells across sorting strategies, diets, and SMC-Klf4 genotypes, and provides a foundation for targeted differential expression and pathway analyses.

Keywords: Adipose tissue, single cell RNA sequencing, obesity metabolic syndrome, inflammation macrophages

Simon Ambrozak (University of Maryland), Ulysse McConnell (ETH Zurich), Bhargav Srinivasan (University of Maryland), Burak Ozkan (Bilkent University) and Can Firtina (University of Maryland).
CERN: Correcting Errors in Raw Nanopore Signals Using Hidden Markov Models.

Abstract. Nanopore sequencing can read substantially longer sequences of nucleic acid molecules than other sequencing methods, which has led to advances in genomic analysis such as the gapless human genome assembly. By analyzing the raw electrical signal reads that nanopore sequencing generates from molecules, existing works can map these reads without translating them into DNA characters (i.e., basecalling), allowing for quick and efficient analysis of sequencing data. However, raw signals often contain errors due to noise and mistakes when processing them, which limits the overall accuracy of raw signal analysis.

Our goal in this work is to detect and correct errors in raw signals to improve the accuracy of raw signal analyses. To this end, we propose CERN, a mechanism that trains and utilizes a Hidden Markov Model (HMM) to accurately correct signal

errors. Our extensive evaluation on various datasets including E. coli, Fruit Fly, and Human genomes shows that CERN 1) consistently improves the overall mapping accuracy of the underlying raw signal analysis tools, 2) minimizes the burden on segmentation algorithm optimization with newer nanopore chemistries, and 3) functions without causing substantial computational overhead. We conclude that CERN provides an effective mechanism to systematically identify and correct the errors in raw nanopore signals before further analysis, which can enable the development of a new class of error correction mechanisms purely designed for raw nanopore signals.

Source Code: CERN is available at https://github.com/STORMgroup/CERN. We also provide the scripts to fully reproduce our results on our GitHub page.

Keywords: raw signals, nanopore sequencing, error correction, hidden markov models, machine learning

Aidan Jan (Carnegie Mellon University) and Guillaume Marçais (Carnegie Mellon University). *Hypercube structure of local decycling set operations.*

Abstract. The de Bruijn graph provides a one-to-one mapping between strings and graph walks, connecting string algorithms and graph theory. This connection serves as a foundation for many algorithmic methods in computational biology, including genome assembly, string matching, and sequence sketching.

Cycles in a de Bruijn graph correspond to repetitive sequences, and their structure constrains the efficiency of k-mer-based sketching methods. We study Minimum Decycling Sets (MDS), smallest sets of k-mers that intersect every cycle, equivalently appearing in any sufficiently long sequence.

Recent work introduced F-moves, local operations on MDSes that exchange selected k-mers for others while maintaining the decycling set property. This enables traversal over the space of decycling sets that permits optimization of underlying sketching methods with window guarantees.

We modeled the network of MDSes joined by F-moves as a graph and identified a previously unreported structural organization. The graph admits substructures isomorphic to n-dimensional hypercubes, where n is the maximum number of independently applicable F-moves on a given MDS. This follows from the locality and commutativity of F-moves, which allow them to be executed in any order. This structure provides a new perspective on the organization of MDSes and suggests new methods for analyzing and constructing decycling sets with desirable properties.

Keywords: de Bruijn graph, sketching, graph decycling

Jae Yeon Hwang (University of Louisville), Zhenhua Shang (University of Louisville), Abdelrahman Mahmoud (University of Louisville) and Juw Won Park (University of Louisville). *Systematic Identification of Translated Circular RNAs through Integrated RNA-seq and Ribo-seq Analysis.*

Abstract. Circular RNAs (circRNAs) are covalently closed RNA molecules generated through back-splicing and are increasingly recognized as important regulators of gene expression. Recent studies suggest that a subset of circRNAs can be translated into functional peptides; however, systematic identification of actively translated circRNAs remains challenging due to the lack of computational frameworks that integrate transcriptomic and ribosome profiling data.

We developed an automated and modular pipeline to detect ribosome-associated circRNAs by integrating RNA-seq and Ribo-seq datasets. The workflow identifies circRNAs from rRNA-depleted RNA-seq or circRNA-seq data and constructs back-splice junction (BSJ) reference sequences. Ribosome-protected fragments from Ribo-seq are aligned to both the genome and BSJ references to detect junction-spanning reads that provide direct evidence of circRNA translation. To ensure high-confidence results, the pipeline applies standardized filtering to retain uniquely mapped reads and reduce false positives commonly observed in junction-based analyses.

Application of the pipeline to publicly available datasets identifies hundreds to thousands of circRNA candidates per dataset, with a subset showing strong translational signatures supported by BSJ-spanning ribosome footprints. Compared to naïve junction-mapping approaches, the pipeline reduces ambiguous mappings and improves specificity while maintaining high sensitivity for detecting candidate coding circRNAs. The approach provides a reproducible framework for screening translated circRNAs across diverse biological conditions.

This work offers a scalable computational solution for studying circRNA translation and facilitates deeper investigation into the functional roles of circRNAs and their contributions to proteomic diversity.

Keywords: Circular RNA, RNA-seq, Ribo-seq, Ribosome profiling, Back-splice junction, Translation detection

Howard Yang (National Cancer Institute), Anish Thomas (National Cancer Institute), Maxwell Lee (National Cancer Institute), Lalage Wakefield (National Cancer Institute) and Brett Schroeder (National Cancer Institute). *RNASeq analysis of the hyperprogressive tumor samples.*

Abstract. We analyzed RNASeq data of tumor samples from a clinical trial of bintrafusp alfa, a bifunctional fusion protein targeting 48 PD-L1 and TGF-β in patients with small cell lung cancer (SCLC). Among the evaluable patients, 38% of them experienced hyperprogressive disease (HPD), marked by rapid clinical deterioration. The sequence reads were aligned to the human reference genome (GRCh38/hg38) using the STAR aligner (v2.7.8a). Gene-level read quantification was performed using the featureCounts function from the Rsubread package. Differential gene expression analysis was conducted using the DESeq2 R package on raw count data comparing HPD samples and the other tumor samples. Functional annotation and biological pathway enrichment analyses were conducted using Gene Set Enrichment Analysis (GSEA v4.3.2).

We applied 16 gene signatures in literature to analyze the samples and found that two signatures cell_cycle_progression and replication_stress were higher in HPD tumors based on after treatment samples. We applied the Cybersort analysis to these samples and found HPD samples were different from the rest samples in these 6 cell types: T cells gamma delta, NK cells resting, Dendritic cells activated, Macrophages M2, T cells CD8, Macrophages M1. We examined 16 stemness and neuroendocrine-ness signatures and found that 13 of them showed higher in HPD samples. These founding revealed some biology characteristics of the HPD samples.

Keywords: RNASeq, hyperprogressive disease, small cell lung cancer

James Lubkowitz (University of Utah) and Kendell Clement (University of Utah). *Transcript-Informed Cell Segmentation for Sequencing-Based Spatial Transcriptomics.*

Abstract. Cell segmentation is central to spatial transcriptomics (ST), as accurate transcript-to-cell assignment impacts downstream analyses. In sequencing-based platforms such as Visium HD, cell segmentation remains challenging due to transcript sparsity, reliance on histological imaging, and spatial binning of transcript capture without single-molecule resolution. Traditional segmentation approaches depend on image-derived features; however, imaging-based ST has shown that incorporating spatial transcript maps improves boundary detection and cell definition.

Here, we present a transcript-informed cell segmentation framework for sequencing-based spatial transcriptomics that integrates image-based nuclei detection with transcript-derived boundary inference. First, we construct a transcriptomic boundary confidence map. At multiple resolutions (e.g., 4 µm, 8 µm, …), we partition the tissue into uniform tiles, aggregate transcripts within each tile, and cluster tiles by expression profiles. These cluster assignments define transcriptional similarity boundaries that capture tissue structure and cell organization. Boundaries across resolutions are combined into a unified map. Next, nuclei are identified and used as seeds for cell reconstruction. Cell boundaries are inferred by assigning each pixel to the nucleus that minimizes a cost function penalizing distance and crossing transcriptomic boundaries. We additionally estimate segmentation uncertainty by perturbing boundary penalties, yielding pixel-level and cell-level confidence scores.

We show that this approach improves boundary delineation and reduces transcript misassignment in Visium HD data, with improved separation of marker gene expression between cells, while providing uncertainty-aware cell segmentations. By combining multiresolution transcriptomic information with image-based priors and explicitly modeling boundary uncertainty, this framework enables more accurate and reliable cell-resolved analysis in sequencing-based spatial transcriptomics.

Keywords: Spatial transcriptomics, Cell segmentation, Visium HD, Transcript-informed segmentation

Zoe Rudnick (Johns Hopkins University, Center for Computational Biology, Baltimore, MD), Ales Varabyou (Johns Hopkins University, Center for Computational Biology, Baltimore, MD), Rob Patro (University of Maryland, Department of Computer Science, College Park, MD) and Mihaela Pertea (Johns Hopkins University, Center for Computational Biology, Baltimore, MD). *Bramble: projection of spliced genomic alignments into transcriptomic space for improved transcript quantification estimation.*

Abstract. Accurate transcript quantification is challenged by the complexity of alternative splicing and the presence of unannotated transcripts, divergent sequences, and pervasive transcriptional noise within RNA-seq data. Quantification methods such as Salmon and Oarfish traditionally rely on transcriptome mappings, leading to biased transcript assignments that depend on the genome annotation composition. Aligning to the genome with HISAT2 or minimap2 and then projecting the alignments into transcriptomic space may circumvent this problem, because reads best explained by unannotated genomic loci would not be constrained to the space of known transcripts during mapping. Despite the advantages of this approach, an efficient and comprehensive implementation has not yet been realized.

To address this, we developed Bramble, a software tool for projecting spliced genomic alignments into transcriptomic space. Bramble was benchmarked against existing short- and long-read quantification pipelines using simulated data that incorporates realistic transcriptional noise and transcript-level subsampling to mimic annotation incompleteness. We demonstrate that pipelines using Bramble consistently yield superior transcript abundance estimates to those using other methods, and for long-read analysis, where reads contain considerable information about transcript origin via exon-exon junction coverage, Bramble provides particularly substantial improvements. We note that improvements to quantification accuracy become more pronounced as annotation incompleteness increases, indicating the benefit of Bramble to analysis of non-model organisms. Overall, Bramble provides an effective bridge between genome alignment and transcript quantification, improving the accuracy of expression estimates and increasing confidence in downstream findings across a wide range of experimental designs.

Keywords: RNA-seq analysis, Transcript quantification, Genome alignment

Hechen Li (Georgia Institute of Technology) and Xiuwei Zhang (Georgia Institute of Technology).
scModCompare: Disentangling Shared Identity and Modality-Specific Regulatory Programs in Multimodal Single-Cell Data.

Abstract. Paired multimodal single-cell assays promise to trace information flow across the central dogma, yet current computational frameworks (consensus VAEs, graph-based methods, and matrix factorization) prioritize merging modalities into a single representation. This treats biologically meaningful discordance, such as epigenetic priming, post-transcriptional buffering, and RNA velocity, as technical noise to be erased. Existing disentanglement approaches either target batch effects rather than cross-modality biology, rely on linear assumptions ill-suited to developmental manifolds, or validate private spaces only through indirect surrogate metrics.

We present scModCompare, a multimodal variational autoencoder that explicitly partitions latent representations into shared and modality-private subspaces through a two-stage training scheme. Stage 1 trains shared encoders via cross-modal reconstruction with an alignment penalty, mathematically restricting the shared space to mutual information; Stage 2 freezes shared encoders and trains private encoders to capture residual modality-specific variance, preventing collapse between subspaces. We further introduce a Pointwise Log-Likelihood Ratio attribution score that quantifies, for every feature in every cell, the information gain contributed by each latent subspace on a unified information-theoretic scale across heterogeneous data distributions (ZINB, Gaussian, Bernoulli).

Across three benchmark datasets, scModCompare resolves hidden CD4+ T cell subsets and flags doublets and ambient-RNA contamination in CITE-seq bone marrow data; reconstructs trajectory of epigenetic lineage priming in SHARE-seq hair follicle differentiation, decoupling enhancer accessibility from transcriptional activation; and recovers the complete kinetic lifecycle of master regulators from spliced/unspliced RNA without predefined kinetic equations, with private unspliced attribution scores achieving Spearman $\rho = 0.92$ against scVelo-inferred RNA velocity. On a human lymph node spatial multi-omics dataset, the private drivers identified canonical marker genes for each without supervision. By transforming multimodal discordance into an interpretable, quantitative signal at both cell and feature levels, scModCompare enables systematic discovery of temporal lags, modality-exclusive functions, and technical artifacts that integration-focused methods obscure.

Keywords: multi-omics single-cell data, modality difference, multimodal variational auto encoder, disentangle learning

Samuel Coleman IV (University of Utah, Biomedical Informatics) and Kendell Clement (University of Utah, Biomedical Informatics). *offSCope: Primer and Amplicon Level Detection of Mispriming in Targeted Amplicon Single-Cell DNA Sequencing.*

Abstract. High-throughput amplicon-based single-cell DNA sequencing (scDNA) enables allele-specific sequence resolution in individual cells, supporting applications from investigating clonal heterogeneity in cancer to analyzing multiplexed CRISPR edit outcomes. However, these assays are dependent upon multiplexed PCR reactions which can introduce technical artifacts such as off-target mispriming and primer-dimerization. These technical artifacts create background noise which can impact the effective sequence depths of amplicon targets, obscure or hide biological signals, and decrease the reliability of downstream analyses. Without taking steps to measure and interpret these technical artifacts, additional sequencing capacity is wasted and primer libraries for future runs cannot be amended in a data driven manner. Here we introduce offSCope, a tool designed to analyze the quality of scDNA assays, locate misprimed reads, and identify the primers responsible. Using offSCope within a publicly available dataset, we detected mispriming rates between 2-3% on average within each cell. Additionally, the most common type of mispriming event was due to primers initiating reads that overlapped unintended amplicons which, at times, resulted in truncated amplicons that failed to cover targets of interest[KC1.1][SC1.2]. Finally, we observed an inverse relationship between read count and mispriming rate, with cells containing fewer reads exhibiting higher levels of mispriming, potentially indicating a role for the mispriming rate as a useful quality control metric in scDNA data. By enabling researchers to identify problematic primers within their sequencing data, offSCope supports informed primer redesign and assay optimization ultimately improving data quality and biological interpretation of single-cell data.

Keywords: Single-Cell Sequencing, DNA-Sequencing, Amplicon Sequencing

Zahra Zare Jousheghani (University of Maryland) and Rob Patro (University of Maryland). *alevin-fry-lr: Likelihood-Informed UMI Resolution Improves Long-Read Single-Cell RNA-seq Quantification.*

Abstract. Motivation: Long-read single-cell RNA sequencing enables transcript-level analysis within individual cells and offers the potential to improve isoform-aware quantification by capturing substantially more transcript structure than short-read technologies. However, accurate quantification remains challenging because single-cell data are sparse and noisy. Many existing single-cell quantification frameworks were developed primarily for short-read data and do not fully exploit the alignment-level information available in long reads. Likewise, some methods developed specifically for long-read data forgo the highly-effective graph-based UMI resolution strategies developed for short-read single-cell data.

Results: We present alevin-fry-lr, a long-read extension of the alevin-fry framework for single-cell RNA-seq quantification. Our method extends the parsimony-based UMI deduplication and quantification model by incorporating alignment-informed evidence derived from the oarfish bulk long-read model. Specifically, we assign alignment-score-based probabilities to candidate transcript alignments and use transcript-specific likelihood scores to guide graph-based UMI resolution that uses a dynamic program to derive a maximum likelihood decomposition of the graph, followed by EM-based abundance estimation for remaining ambiguity. We evaluated alevin-fry-lr on simulated and experimental long-read single-cell datasets generated using ONT and PacBio technologies. Simulated datasets were generated with TKSM, and experimental datasets were obtained from LongBench, a cross-platform benchmark comprising 10X Chromium single-cell libraries from eight lung cancer cell lines sequenced with ONT and PacBio. Compared with Isosceles, kallisto, and bambu, alevin-fry-lr achieved the highest Spearman and Pearson correlations and the lowest NRMSE across cells, while also being one of the most time and memory frugal methods.

Keywords: long-read, single-cell, RNA-seq, transcript quantification, UMI deduplication, graph-based parsimony, probabilistic modeling, ONT, PacBio

Haodong Chen (Sentieon Inc.), Donald Freed (Sentieon Inc.), Jinnan Hu (Sentieon Inc.), Hong Chen (Sentieon Inc.), Hanying Feng (Sentieon Inc.), Zhipan Li (Sentieon Inc.), Brendan Gallagher (Sentieon Inc.) and Luoqi Chen (Sentieon Inc.). *Accurate and comprehensive germline variant calling leveraging public pangenomes.*

Abstract. Pangenome references overcome limitations of the traditional linear reference genome by incorporating population-specific haplotypes into graph-based frameworks, but their practical adoption has been constrained by high computational costs and limited downstream variant-calling support. Here we present the Sentieon DNAscope Pangenome pipeline, which leverages public pangenome graphs to construct personalized haplotype references, followed by realignment to correct systematic mapping errors and machine learning-based variant callers to detect SNPs/indels, SVs, and CNVs from short-read sequencing data. Extensive benchmarking on Genome in a Bottle (GIAB) datasets demonstrates state-of-the-art accuracy across multiple short-read sequencing platforms. Benchmarked against GIAB truth sets, the pipeline achieved F1 scores of 99.89%/99.78% (SNP/indel) on 30× Illumina data with comparable accuracy on two additional short-read platforms. SV performance reached an F-measure of 95.95% on HG002, a dramatic improvement over linear-reference approaches (Manta: 48%), with consistent results on independent CMRG benchmarks confirming generalizability. CNV evaluation yielded an overall F-measure of 90.16%, with length-stratified analysis showing F1 scores exceeding 95% for events larger than 3 kbp, demonstrating high sensitivity for clinically relevant larger events. Starting from pre-aligned whole-genome input, the pipeline completes in ~80 core-hours (~$4.30 on-demand), making pangenome-informed variant calling economically viable at the population scale.

Keywords: pangenome, germline variant calling, structural variant, copy number variant, machine learning, short-read sequencing, benchmarking

Bhargav Srinivasan (University of Maryland), Simon Ambrozak (University of Maryland) and Can Firtina (University of Maryland). *Label-Free Detection of Epigenetic Modifications from Raw Nanopore Signals.*

Abstract. Epigenetic modifications such as 5-methylcytosine (5mC), 5-hydroxymethylcystosine (5hmC), and N6-methyladenine (6mA) play crucial roles in gene regulation, biological signaling, and other downstream biological processes. Profiling and detecting these modifications at the single-base resolution is required to understand a full scope of genomic function across contexts beyond just the canonical base pair space. While nanopore sequencing offers a powerful platform for this purpose, existing epigenetic modification tools are limited to a small set of modifications and rely on costly machine learning models or articulate algorithms for each set of modifications.

We present ProMURS (Profiling Modifications Using Raw Signals), a label-free framework for epigenetic modification detection directly from raw nanopore signals. ProMURS constructs a custom event pileup by aligning signal with a shared subsequence across reads and then performs downstream analyses to differentiate between modified and unmodified reads. These approaches entail a K-Means clustering on vectorized pileup values, a Principal Component Analysis based separation between modification types, and Dynamic Time Warping cost based error function as a differentiation between modified and unmodified reads.

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On Oxford Nanopore Technologies Open Source Modification Benchmarking Dataset, ProMURS successfully differentiates unmodified and modified reads from 5mC, 5hmC, and 6mA modifications. Results show clear differentiation, with a PCA accuracy of up to 96% determined by 1-NN classification. ProMURS key contribution to the field is a modification pipeline that does not require labeled training data or prior modification knowledge.

Keywords: Nanopore Sequencing, DNA Modification, Unsupervised Learning

Muhammad Atif Sikandar (Institute of Pharmcology and Toxicology, University Medical Center Gottingen, Germany), Jingyu Wang (Institute of Pharmcology and Toxicology, University Medical Center Gottingen), Aagash Nedunchezhian (Institute of Pharmcology and Toxicology, University Medical Center Gottingen), Imran Ullah (Institute of Pharmcology and Toxicology, University Medical Center Gottingen), Wolfram-Hubertus Zimmermann (Institute of Pharmcology and Toxicology, University Medical Center Gottingen) and Olaf Bergmann (Institute of Pharmcology and Toxicology, University Medical Center Gottingen, Karolinska Institutet, Sweden). *Multi-omic Profiling Reveals Cardiac Microenvironments and Cellular Diversity in iPSC-derived Cardiac Bioengineered Heart Muscle Organoids.*

Abstract. Human induced pluripotent stem cell (iPSC)-derived cardiac bioengineered heart muscle (BHM) organoids present a promising platform for recapitulating key aspects of cardiac development and disease in vitro. In this study, we combined temporal bulk RNA sequencing with high-resolution spatial transcriptomic analyses to systematically characterise the differentiation trajectory and structural maturation of BHM organoids. Integrated analysis revealed a heart-like architecture characterised by the spatial distribution of cardiomyocytes, fibroblasts, endothelial cells, epicardial cells, and other support cell types. Furthermore, spatial mapping revealed the organised arrangement of these cell types within in vivo-like microenvironments. Preliminary hypoxia experiments also highlight the potential of BHM organoids as a model for mechanistic studies of cardiac pathology and drug screening. Overall, our findings establish iPSC-derived BHM organoids as a versatile and functionally competent system for investigating human heart development, disease mechanisms and therapeutic evaluation.

Keywords: Bioengineered heart muscle, Spatial transcriptomics, Bulk RNA-seq, Cardiac systems biology, Hypoxia, Transcriptomic integration, Cardiac remodeling

Adam Gudys (Silesian University of Technology), Andrzej Zieleziński (Adam Mickiewicz University) and Sebastian Deorowicz (Silesian University of Technology). *Kmer-db: latest improvements in k-mer-based sequence indexing and comparison.*

Abstract. The exponential growth of sequence databases results in a need for fast, alignment-free sequence comparison algorithms. Most existing methods employ k-mers, often combining them with MinHash sketching to reduce representation size (e.g., Mash, Dashing). Kmer-db, our tool for k-mer-based sequence analysis, has confirmed its usability in many applications, with Vclust, a state-of-the-art package for large-scale viral genomes clustering, being the paramount example.

Here we present a new version of Kmer-db. The algorithmic improvements result in substantial reduction of execution times, particularly on multicore architectures. An automatic split of data into batches based on sequence count or memory requirements, removes practical barriers for the dataset sizes that can be processed. The support of protein sequences with an optional alphabet reduction further increases the usability. Kmer-db database stores by default all the k-mers, thus can be accurately queried for their presence in the sequences. This is an advantage over data structures with a non-zero false positive rate employed by some packages (BIGSI, COBS). However, when one is interested only in identity estimation, the MinHash-based filtering is also available.

When evaluated across different modes and configurations, Kmer-db was 70% to 5 times faster than its predecessor. For example, the comparison of over 15 million viral genomes from IMG/VR v4 repository at 0.95 nucleotide identity level required less than half an hour. We believe that superior performance combined with multiple usability features make Kmer-db ready for future challenges in genomics and proteomics.

Keywords: sequence comparison, nucleotide identity, k-mers, MinHash, metagenomics, high-throughput sequencing

Federica Bichicchi (National Institutes of Health), Natalie Atyeo (National Institutes of Health), Danny Yi (National Institutes of Health) and Jay Chiorini (National Institutes of Health). *Design and validation of a spatial transcriptomics viral panel for virome profiling in Sjögren's Syndrome.*

Abstract. Sjögren's Syndrome (SjD) is a chronic autoimmune disorder affecting salivary and lacrimal glands, leading to severe dryness and systemic complications. While its exact cause remains unknown, genetic predisposition and environmental triggers are thought to contribute to disease onset and progression. Several viruses are known to infect salivary gland cells, and emerging evidence suggests that persistent infections may drive immune dysregulation and chronic inflammation in SjD. However, the specific viruses involved and their impact on disease pathology remain unclear. To address this, we developed and validated a spatial transcriptomics viral probe panel. We implemented a computational pipeline to retrieve viral genomes, perform multiple sequence alignment, and quantify sequence conservation using normalized Shannon entropy. Conserved regions were selected to guide probe design across genomes with varying variability, resulting in a panel of 258 viral targets. For experimental validation, representative viral fragments were cloned into plasmids and embedded in tissue microarrays (TMAs), enabling controlled benchmarking of probe performance within a Xenium spatial transcriptomics framework. Control experiments demonstrated robust detection of viral and host transcripts with consistent signal across replicated cores. Integration of probe-level expression with known ground truth enabled quantitative assessment of sensitivity and specificity, revealing strong performance for targets and identifying potential cross-reactivity in variable genomes. This study establishes a scalable framework for the design and validation of spatial viral panels, supporting accurate virome profiling in salivary gland tissues. Future work will apply this approach to SjD patient samples to investigate virus–host interactions and their role in disease pathogenesis.

Keywords: Sjogren's Syndrome, SjD, Virus, Spatial Transcriptomics, Xenium

Thomas Brooks (University of Pennsylvania). *Empirical Bayes inference in circadian omics.*

Abstract. Empirical Bayes has widespread use in omics as part of packages such as DESeq2 and limma. However, most users only make use of its variance shrinkage properties which enable more robust p-values but do not affect effect-size estimates. Here, we introduce a software package that enables flexible empirical Bayes models. We demonstrate this by applications to complex circadian datasets which have time courses in multiple conditions. Importantly, this approach addresses a need for high-level summaries of differential expression or differential rhythms, improving on the widespread, but questionable, use of Venn diagrams. Our system improves the interpretability of results over existing empirical Bayes frameworks and is tailored to be sufficiently flexible to accommodate typical complex study designs.

Keywords: empirical Bayes, omics, RNA-seq, effect size shrinkage, software

Reema Singh (Vaccine and Infectious Disease Organization, University of Saskatchewan, Saskatoon, SK, Canada). *CdMEC-A Enables High-Throughput One Health Analysis of ARG–MGE Genomic Context in Clostridioides difficile.*

Abstract. The spread of antimicrobial resistance (AMR) via mobile genetic elements (MGEs) represents a critical One Health challenge, yet scalable computational approaches for quantifying spatial relationships between antimicrobial resistance genes (ARGs) and MGEs remain limited.

CdMEC-A is a high-throughput Python-based pipeline for automated detection, spatial quantification, and risk stratification of ARG–MGE associations. It computes genomic distances between antimicrobial resistance genes and MGEs, enabling systematic assessment of mobilization potential across large-scale genomic datasets.

The pipeline was validated using the Clostridioides difficile 630 reference genome and benchmarked on 495 RefSeq genomes alongside 448 isolates from human, porcine, and environmental sources. CdMEC-A achieved >99.5% feature mapping accuracy and demonstrated highly efficient parallel performance, processing over 12,000 genomes per day on a 16-core system.

Application across One Health datasets revealed widespread ARG–MGE coupling, with 9.8% of ARGs embedded within MGEs. Cross-host comparisons highlighted conserved and host-associated patterns of resistance organization across human, porcine, and environmental reservoirs.

CdMEC-A provides a scalable and reproducible framework for mobilome-aware AMR surveillance, enabling prioritization of high-risk genetic configurations and supporting large-scale comparative genomic analysis.

Keywords: Antimicrobial resistance, Mobile genetic elements, One Health, Clostridioides difficile

Jianing Yang (University of Pennsylvania), Gregory Grant (University of Pennsylvania) and Thomas Brooks (University of Pennsylvania). *Generating correlated data for omics simulation.*

Abstract. Simulation of realistic omics data is a key input for benchmarking studies that help users obtain optimal computational pipelines. Omics data involves large numbers of measured features on each sample and these measures are generally correlated with each other. However, simulation too often ignores these correlations. To alleviate this, we describe and implement three approaches for generating omics-scale data with correlated measures which mimic real datasets. These approaches are all based on a Gaussian copula approach with a covariance matrix that decomposes into a diagonal part and a low-rank part, which allows for extremely efficient simulation. We use these approaches to demonstrate the importance of including correlation in two benchmarking applications. First, we show that variance of results from the popular DESeq2 method increases when dependence is included. Second, we demonstrate that CYCLOPS, a method for inferring circadian time of collection from transcriptomics, improves in performance when given gene-gene dependencies in some circumstances.

Keywords: simulation, correlation, benchmarking, transcriptomics

Antonijo Mrcela (University of Pennsylvania), Thomas Brooks (University of Pennsylvania), Nicholas Lahens (University of Pennsylvania), Jianing Yang (University of Pennsylvania), Souparna Purohit (University of Pennsylvania), Amruta Naik (University of Pennsylvania), Emanuela Ricciotti (University of Pennsylvania), Shaon Sengupta (Children's Hospital of Philadelphia), Peter Choi (Children's Hospital of Philadelphia) and Gregory Grant (University of Pennsylvania). *A systematic evaluation of the sources of non-uniform coverage in RNA-Seq.*

Abstract. Extreme non-uniformity of transcript coverage remains one of the central unresolved technical artifacts in short-read RNA-Seq. These large-scale coverage biases substantially limit isoform-level quantification and impact a wide range of downstream analyses, including alternative splicing detection, transcript abundance estimation, benchmarking studies, and realistic simulation of RNA-Seq data. Despite extensive prior work, the dominant sources of this non-uniformity remain unclear.

We performed a systematic evaluation of candidate sources of RNA-Seq coverage bias using both public datasets and newly generated libraries with controlled variation in ribosomal depletion strategy, fragmentation conditions, PCR ramp rate, fragment length, and PCR amplification. Coverage profiles were highly reproducible across technical replicates and reasonably consistent across biological replicates, demonstrating that the observed patterns are not primarily driven by stochastic fragment sampling.

We show that several previously proposed explanations fail to substantially improve uniformity, including reduced PCR ramp rate, fewer PCR cycles, omission of ribosomal depletion, shorter fragment lengths, and use of thermostable reverse transcriptases. In contrast, PCR duplicate rates correlated with local coverage in a library-dependent manner, implicating amplification efficiency as an important contributor to coverage variation. However, decreasing PCR cycle number alone did not substantially improve uniformity, suggesting that molecular abundance and saturation-related effects may be more important than cycle count itself.

Overall, our results demonstrate that RNA-Seq coverage non-uniformity arises from multiple interacting technical factors rather than a single dominant mechanism, with important implications for transcript quantification, benchmarking methodology, and RNA-Seq simulation frameworks.

Keywords: Coverage bias, Isoform-level quantification, Alternative splicing detection, RNA-Seq simulation, PCR

Nika Mansouri Ghiasi (ETH Zurich), Talu Güloglu (ETH Zurich), Harun Mustafa (ETH Zurich and Johns Hopkins University), Can Firtina (University of Maryland), Konstantina Koliogeorgi (ETH Zurich), Konstantinos Kanellopoulos (ETH Zurich), Haiyu Mao (King's College London), Rakesh Nadig (ETH Zurich), Mohammad Sadrosadati (ETH Zurich), Jisung Park (Pohang University of Science and Technology (POSTECH)) and Onur Mutlu (ETH Zurich). *SAGe: A Lightweight Algorithm-Architecture Co-Design for Mitigating the Data Preparation Bottleneck in Large-Scale Genome Sequence Analysis.*

Abstract. Genome sequence analysis drives advances in many critical medical and biotechnological fields. Given its importance and the exponentially growing volumes of genomic sequence data, there are extensive efforts to accelerate genome sequence analysis. In this work, we demonstrate a major bottleneck that greatly limits and diminishes the benefits of state-of-the-art genome sequence analysis accelerators: the data preparation bottleneck, where genomic sequence data is stored in compressed form and needs to be first decompressed and formatted before an accelerator can operate on it. To mitigate this bottleneck, we propose SAGe, an algorithm-architecture co-design for highly-compressed storage and high-performance access of large-scale genomic sequence data. The key challenge is to improve data preparation performance while maintaining high compression ratios (comparable to genomic-specific compression algorithms) at low hardware cost. We address this challenge by leveraging key properties of genomic datasets to co-design (i) a lossless (de)compression algorithm, (ii) hardware that decompresses data with lightweight operations and efficient streaming accesses, (iii) storage data layout, and (iv) interface commands to access data. SAGe is highly versatile, as it supports datasets from different sequencing technologies and species. Due to its lightweight design, SAGe can be seamlessly integrated with a broad range of hardware accelerators for genome sequence analysis to mitigate their data preparation bottlenecks. Our results demonstrate that SAGe improves the average end-to-end performance and energy efficiency of two state-of-the-art genome sequence analysis accelerators by 3.0×–32.1× and 13.0×–34.0×, respectively, compared to when the accelerators rely on state-of-the-art software and hardware decompression tools.

Keywords: Genome sequence analysis acceleration, Compression and Decompression, High performance, Energy efficiency

Nika Mansouri Ghiasi (ETH Zurich), Harun Mustafa (Johns Hopkins University), Talu Güloglu (ETH Zurich), Rakesh Nadig (ETH Zurich), Konstantinos Kanellopoulos (ETH Zurich), Susana Rebolledo Ruiz (ETH Zurich, University of Cantabria), Marc Rautmann (ETH Zurich), Furkan Eris (ETH Zurich), Mohammad Sadrosadati (ETH Zurich), Jisung Park (Pohang University of Science and Technology (POSTECH)) and Onur Mutlu (ETH Zurich). *GRAINS: Enabling High-Performance and Low-Cost Graph-Based Genome Analysis via Storage-Aware Algorithm-Architecture Co-Design.*

Abstract. Graph-based representations of genome sequences have emerged as a powerful approach for representing massive genomic databases expressively and efficiently. However, analysis on large-scale genome graphs incurs significant data movement overhead from the storage system due to accessing large amounts of low-reuse data. Processing data directly inside the storage device, where data originally resides, can fundamentally mitigate this overhead. However, no existing tool for graph-based genome analysis can be efficiently used inside storage due to the limited hardware resources in modern SSDs.

We propose GRAINS, the first system for analysis with large-scale genome graphs in storage. Through our detailed examination of typical analysis pipelines that operate on genome graphs, we perform storage-aware algorithm-architecture co-design to (i) make these pipelines more storage-friendly and (ii) further improve performance, energy-efficiency, and cost via in-storage and in-flash processing. GRAINS has three key aspects. First, we propose a new batching technique and execution flow, based on unique features of genome graphs, that reduces random accesses. Second, via in-flash and in-storage processing, we avoid transferring low-reuse or unused flash pages, preventing bandwidth waste. Third, to leverage the full parallelism of flash dies, we design a lightweight scheduling technique, enabled by re-purposing the existing SSD structures. GRAINS's design is versatile and flexible as it supports key operations on genome graphs and can be integrated into various analysis pipelines. GRAINS provides 2.7×–47.8× speedup (4.4×–31.6× energy reduction) over the state-of-the-art software baselines, and 1.5×–17.0× speedup (3.1×–20.7× energy reduction) over a hardware-accelerated baseline.

Keywords: Graph-Based Genome Analysis, Storage-Aware Algorithms, Algorithm-Architecture Co-Design, Storage Systems, Storage-Centric Computing, High Performance, Energy Efficient, Low Cost

Yifei Wang (Yale School of Medicine, BIDS) and Haoyu Cheng (Yale School of Medicine, BIDS). *PopPhaser: Joint Unitig-Based and Population-Level Haplotype Phasing.*

Abstract. Accurate diploid haplotype phasing is essential for population genetics, disease association studies, and genome assembly. When Hi-C data is unavailable, phasing quality depends on how efficiently read-level and population-level information can be jointly exploited. Existing tools such as WhatsHap and SHAPEIT4 address these two sources separately: WhatsHap constructs phase blocks from sequencing reads, and SHAPEIT4 scaffolds those blocks using linkage disequilibrium (LD) from a reference panel. Because they operate sequentially, neither tool can jointly optimize the weighted Minimum Error Correction (wMEC) objective, leaving the two sources only partially integrated.

We present PopPhaser, which jointly exploits unitig-based assembly-level and population-level evidence within a single phasing framework. PopPhaser uses error-corrected unitigs from a hifiasm de novo assembly as phasing fragments. Unitigs are substantially longer than raw reads and carry much lower local coverage, making exact wMEC inference tractable and reducing orientation ambiguity at biallelic heterozygous variant sites. Population LD is incorporated at two stages: 1) a PBWT-restricted Li-Stephens transition penalty is embedded directly into the wMEC dynamic program, gated by an ambiguity score so that population information guides orientation only when unitig evidence is insufficient; 2) SHAPEIT4 with phased block scaffolds fills inter-block gaps and refines block orientations using the full reference panel.

Evaluated on HG002 HiFi sequencing data, PopPhaser substantially outperforms Hifiasm dual assembly across all phasing metrics. Switch error rate improves from 0.7557%/0.7525% to 0.6518%/0.6105%, Hamming error rate drops from 23.7099%/22.3092% to 20.1329%/19.1529%, and phased block continuity improves markedly with Phased Block N50 from ~1.1 Mb to ~3.9 Mb compared to hifiasm dual assembly. These results demonstrate that PopPhaser improves both phasing accuracy and continuity, providing a practical solution for haplotype phasing in diploid genomes without Hi-C data.

Keywords: haplotype phasing, genome assembly, weighted Minimum Error Correction, long-read sequencing, population genetics, linkage disequilibrium, HiFi sequencing

Zhezheng Xander Song (Carnegie Mellon University) and Carl Kingsford (Carnegie Mellon University).
TIERIA: Quantum Best-Pair Search for Biological Sequences via k-mer Distribution Statistics.

Abstract. Finding highly similar pairs of biological sequences is a fundamental task in bioinformatics, with applications in redundancy reduction, clustering, database cleaning, comparative genomics, and metagenomic analysis.

While classical tools often avoid exhaustive all-pairs comparison through heuristics, sketching, or threshold-based clustering, exact closest-pair search remains naturally modeled as an unstructured maximum-finding problem for many sequence similarity measures.

We present TIERIA, a quantum framework for alignment-free closest-pair search under a distributional \(k\)-mer similarity measure.

Each sequence is represented by the empirical distribution of its observed \(k\)-mers and encoded as a quantum state whose amplitudes are the square roots of these frequencies.

Under this encoding, the squared overlap between two sequence states equals the squared Bhattacharyya coefficient of their \(k\)-mer distributions.

TIERIA estimates this similarity using a SWAP-test-based quantum amplitude estimation procedure and embeds the resulting threshold-comparison oracle into a D\"urr--H{\o}yer maximum-finding framework with a BBHT-style search schedule.

Under an idealized qRAM-style data-access assumption, TIERIA performs closest-pair search in

\widetilde O\!\left(\frac{NL}{\varepsilon}\right)

time, reducing the leading dependence on the number of sequences from quadratic to linear compared with the classical exhaustive baseline \(O(N^2L)\).

We also implement a proof-of-concept version in Q\#, which does not assume qRAM and instead uses sparse state-preparation schedules with explicit indexed loading.

Including this loading cost, the implementation runs in

\widetilde O\!\left(\frac{N^2L}{\varepsilon}\right),

matching the classical exhaustive scaling while validating the circuit-level framework.

Our results provide a concrete quantum framework for studying alignment-free sequence comparison and highlight the role of efficient coherent data access in realizing asymptotic quantum advantage for omics search problems.

Keywords: quantum algorithms, closest-pair search, k-mer statistics, Bhattacharyya coefficient, quantum amplitude estimation, qRAM